

Explicit Boundedness, Exact Upper Elimination, and Lower-Range Endpoint States for the Polynomials $Q_{a,b}$

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Abstract

For

$$Q_{a,b}(x) = \frac{ax^{a+b} - (a+b)x^a + b}{(x^{\gcd(a,b)} - 1)^2} \quad (a, b \geq 1),$$

we study the conjecture that reciprocal factor packages attached to distinct unordered pairs are pairwise coprime over $\mathbb{Q}$. The preceding revision proved, by an exact integer computation, that every normalized counterexample satisfies

$$\max\{m, n, p, q\} \leq 6\,816\,241.$$

The present revision develops the lower-range algebra needed to turn that remaining box into an auditable finite-state computation.

The principal new geometric input is a reduced power-correspondence bound. If selected primitive orientations have shapes (a, b) , (c, e) and coprime scales r, s , then every common irreducible factor of degree d satisfies

$$d \leq \frac{rs(ae + bc)}{\gcd(rb, se) \gcd(ra, sc)}.$$

This contains the former same-shape estimate $d \leq 2rs$. We retain the actual extraction degrees

$$t = [\mathbb{Q}(\alpha) : \mathbb{Q}(\alpha^r)], \quad u = [\mathbb{Q}(\alpha) : \mathbb{Q}(\alpha^s)]$$

instead of assuming full Kummer degree. Thus, with e_1, e_2 the two primitive factor degrees,

$$d = te_1 = ue_2, \quad d \leq e_1 e_2.$$

Scaled norm identities give exact endpoint divisibilities and slope matching. If both endpoint coefficients of the common minimal polynomial are nonunits, these estimates force $D \leq 37\,799$. Above 50 000, at most one coefficient is nonunit and its absolute value is at most 12.

A deficient scale prime is also localized without assuming full Kummer degree. In the unit case it is a good prime and forces a small primitive factor degree. In the nonunit case, the only branch not covered by that small-degree alternative is an endpoint packet at a prime in $\{2, 3, 5, 7\}$, with exact norm, slope, and packet constraints. These conditions define a finite certificate object for every lower-range branch.

The accompanying bundle retains the independently audited upper computation and adds exact constant certificates, a lower-state schema, reference checkers, modular-gcd certificate tooling, and an explicit proof-status boundary. The new reductions are proved in the text, but the production endpoint-state enumeration has not yet received an independent coverage audit. Accordingly this revision is a candidate route to a complete computer-assisted proof, not yet a claim that the full conjecture has been proved.

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1 Statement, strategic choice, and principal results

The three sufficient targets in the motivating problem are:

- (a) every unordered unequal pair has a globally private irreducible factor;
- (b) reciprocal packages of distinct unordered pairs are pairwise coprime;
- (c) every $Q_{a,b}$, $a \neq b$, is irreducible over $\mathbb{Q}$.

The primitive subcase of (c) is Borisov’s irreducibility problem for the abc -polynomials [1]; the general case additionally contains the composition problem

$$Q_{rA,rB}(x) = rQ_{A,B}(x^r).$$

Target (b) remains the cleanest uniform statement. The new self-power theorem removes every possible collision having even one irreducible primitive base, so the unresolved core now lies entirely in the reducible–reducible family.

Write

$$P_{a,b}(x) = Q_{a,b}(x)Q_{b,a}(x)$$

for the reciprocal package product. Call the unordered pair $\{a, b\}$ *globally isolated* if $P_{a,b}$ is coprime to every $P_{c,d}$ with $\{c, d\} \neq \{a, b\}$.

Theorem 1.1 (Summary).

(i) *Every noncyclotomic irreducible reciprocal factor orbit occurs in only finitely many packages. Hence the package-sharing graph is locally finite.*

(ii) *For every primitive unequal pair (a, b) , including boundary pairs,*

$$Q_{a,b} \mid Q_{c,d} \implies (c, d) = (a, b).$$

(iii) *Let A, B be coprime and unequal, and suppose $Q_{A,B}$ is irreducible. If β is a root, then*

$$\beta \notin \mathbb{Q}(\beta)^p \quad \text{for every prime } p.$$

Every lift $\{rA, rB\}$ is therefore globally isolated.

- (iv) If $I(X)$ counts unordered unequal pairs of total at most X whose packages are not globally isolated, then

$$I(X) = O\left(X^{20/11} \log X\right).$$

Thus a density-one set of package vertices is globally isolated from the entire infinite family.

- (v) If an irreducible factor of degree e divides a primitive $Q_{A,B}$, then

$$\text{rad}(A+B) \mid e(e+2).$$

For every prime $p \mid A+B$, including $p=2$ with arbitrary $v_2(A+B)$, a factor root is not a p -th power in its own field.

- (vi) If $p^\kappa \parallel B$ and a degree- e factor root is a p -th power in its own field, then

$$e \leq A, \quad pA \mid e\kappa,$$

for every prime p , with the reciprocal statement when $p^\kappa \parallel A$. Thus the only local power room at a bad endpoint is inside its nonunit block, and it requires $\kappa \geq p$.

- (vii) If $N = qk$ with q odd prime and $Q_{A,B} = \prod_i h_i$, then the factor degrees satisfy the exact partition identities of Corollary 15.11. In particular, if $q > k+1$, exactly one factor has degree congruent to $-2 \pmod{q}$ and there are at most k factors. As a consequence, primitive total q or $2q$, with q an odd prime, always gives an irreducible $Q_{A,B}$.

- (viii) Every normalized four-index counterexample satisfies

$$\max\{m, n, p, q\} \leq 175\,394\,637.$$

If the maximum is at least 6 816 242, the primitive minimal polynomial of a common root is monic with constant coefficient ± 1 .

- (ix) An exact integer computation eliminates that entire unit-normalized range. Consequently every normalized counterexample satisfies

$$\boxed{\max\{m, n, p, q\} \leq 6\,816\,241.}$$

The computation is reduced to 761 same-shape rows, all excluded by the incompatible bounds $d \leq 2rs \leq 38\,364$ and $d \geq 48\,688$.

- (x) If a primitive irreducible polynomial h , of degree e and Mahler measure $M(h)$, divides selected orientations of two primitive packages, then the radical of the product of their six additive-triple entries is at most

$$2eM(h)^{4-1/e},$$

and at most $2eM(h)^{2-1/e}$ in the monic, unit-constant case. The leading and constant coefficients also satisfy the four endpoint divisibilities (17.24), and their prime divisors force the exact slope matching (17.25)–(17.26).

The density estimate uses Borisov's quantitative count of exceptional primitive shapes. It does not settle the full conjecture: the remaining proof requirement is the exclusion of compatible proper factors of two reducible primitive abc -polynomials and their compositions. The residual-polynomial splitting theorem removes the former local 2-adic exception, while the height-degree argument and the exact computation in Section 18 eliminate the complete unit-normalized part of the explicit box. What remains is the finite nonunit range $D \leq 6\,816\,241$, for which endpoint-power branching must still be completed.

2 Reciprocal normalization and the collision dictionary

For $0 < m < n$, define

$$H_{m,n}(x) = mx^n - nx^m + (n - m), \quad L_k(x) = \frac{1 - x^k}{k}.$$

Then

$$H_{m,n}(z) = 0 \iff L_m(z) = L_n(z), \quad (2.1)$$

and

$$Q_{a,b}(x) = \frac{H_{a,a+b}(x)}{(x^{\gcd(a,b)} - 1)^2}. \quad (2.2)$$

2.1 Reciprocal packages

Let $g = \gcd(a, b)$ and $N = a + b$. Direct calculation gives

$$x^{N-2g}Q_{a,b}(1/x) = Q_{b,a}(x). \quad (2.3)$$

The monomial in (2.3) is harmless because $Q_{a,b}(0) = b \neq 0$.

Definition 2.1. The reciprocal package of the unordered pair $\{a, b\}$ is

$$\mathcal{P}_{\{a,b\}} = \{Q_{a,b}, Q_{b,a}\},$$

or, equivalently for factor questions, the product $Q_{a,b}Q_{b,a}$.

An edge $\{m, n\}$, $0 < m < n$, in the collision graph corresponds to the unordered package $\{m, n - m\}$. The two orientations of that package are the complementary edges

$$\{m, n\}, \quad \{n - m, n\}. \quad (2.4)$$

2.2 The unit circle is completely removed

The following elementary lemma closes the first logical gap identified in the audit.

Lemma 2.2 (Unit-circle roots). *Let $e = \gcd(m, n)$. If $H_{m,n}(z) = 0$ and $|z| = 1$, then*

$$z^m = z^n = 1, \quad \text{hence} \quad z^e = 1.$$

Conversely, every e -th root of unity is an exactly double root of $H_{m,n}$. Consequently,

$$\frac{H_{m,n}(x)}{(x^e - 1)^2}$$

has no root on the unit circle.

Proof. The collision equation can be written

$$z^m = \left(1 - \frac{m}{n}\right) + \frac{m}{n}z^n.$$

The right side is a strict convex combination of two points of the closed unit disk. Its modulus is one, so equality in the triangle inequality forces $z^n = 1$, and then $z^m = 1$.

If $\zeta^e = 1$, then $H_{m,n}(\zeta) = H'_{m,n}(\zeta) = 0$, while

$$H''_{m,n}(\zeta) = mn(n - m)\zeta^{m-2} \neq 0.$$

Thus the multiplicity is exactly two. These are all the unit-circle roots by the first paragraph. $\square$

These are also all multiple roots. Indeed,

$$H'_{m,n}(x) = mnx^{m-1}(x^{n-m} - 1).$$

If $H_{m,n}(x) = H'_{m,n}(x) = 0$, then $x \neq 0$, $x^{n-m} = 1$, and therefore

$$0 = H_{m,n}(x) = (n - m)(1 - x^m).$$

Thus $x^m = x^n = 1$, so $x^{\gcd(m,n)} = 1$. Consequently every $Q_{a,b}$ is square-free and all of its roots lie off the unit circle. Also

$$H_{m,2m}(x) = m(x^m - 1)^2, \quad Q_{m,m}(x) = m,$$

so an off-unit-circle collision always corresponds to an unequal pair.

2.3 Primitive reduction

If $a = gA$, $b = gB$, and $\gcd(A, B) = 1$, then

$$Q_{a,b}(x) = gQ_{A,B}(x^g). \tag{2.5}$$

Likewise, if a common-root configuration contains the edges $\{m, n\}$ and $\{p, q\}$, and

$$s = \gcd(m, n, p, q), \quad w = z^s,$$

then

$$L_{sk}(z) = \frac{1}{s}L_k(w). \tag{2.6}$$

Thus a minimal counterexample may be normalized by

$$\gcd(m, n, p, q) = 1. \tag{2.7}$$

In that normalization, $\gcd(\gcd(m, n), \gcd(p, q)) = 1$, exactly the coprime-gcd regime occurring in Schinzel's bivariate theorem.

3 No three equal values and additive-triple separation

For fixed z , let $\mathcal{G}_z$ be the graph on the positive integers in which $\{m, n\}$ is an edge precisely when $L_m(z) = L_n(z)$.

Theorem 3.1 (No-three theorem). *If $|z| \neq 1$, there are no three distinct positive integers i, j, k with*

$$L_i(z) = L_j(z) = L_k(z).$$

Consequently, $\mathcal{G}_z$ is a matching.

Proof. Let the common value be A , and put $R = |z|^2 \neq 1$. Then

$$z^r = 1 - rA \quad (r = i, j, k).$$

Consider the real smooth function

$$F(t) = |1 - tA|^2 - R^t.$$

It vanishes at the four distinct real points $0, i, j, k$, whereas

$$F'''(t) = -(\log R)^3 R^t$$

never vanishes. Three applications of Rolle's theorem give a contradiction. $\square$

Because the two complementary edges in (2.4) share their upper endpoint, Theorem 3.1 rules out their simultaneous occurrence at an off-unit-circle root. We may therefore state the exact equivalence that was too early in the first attack.

Proposition 3.2 (Exact collision reformulation). *The reciprocal packages of distinct unordered pairs are pairwise coprime if and only if*

$$|E(\mathcal{G}_z)| \leq 1 \quad \text{for every } z \in \mathbb{C}^\times, |z| \neq 1. \quad (3.1)$$

Proof. If two package products share an irreducible factor, choose a complex root z of that factor. Lemma 2.2 gives $|z| \neq 1$, and an orientation from each package gives two collision edges at z . Conversely, an edge equation has rational coefficients, so a common root of two edges is algebraic; because it is off the unit circle, its minimal polynomial divides the corresponding two Q -polynomials. Finally, two edges representing the same unordered package would be the complementary edges in (2.4), which share their upper endpoint and are forbidden by Theorem 3.1. Thus two distinct graph edges are exactly a common factor between two distinct packages. $\square$

The addition law

$$(u + v)L_{u+v}(z) = uL_u(z) + vz^uL_v(z) \quad (3.2)$$

is fundamental.

Lemma 3.3 (Propagation). *If*

$$L_u = L_{u+v} \quad \text{and} \quad L_v = L_w,$$

then

$$L_u = L_{u+w}.$$

If $w \neq v$, this contradicts Theorem 3.1.

Proof. The first equality and (3.2) give $L_u = z^uL_v$. Hence

$$(u + w)L_{u+w} = uL_u + wz^uL_w = uL_u + wz^uL_v = (u + w)L_u.$$

$\square$

For an unordered pair define its additive triple

$$\mathcal{T}(a, b) = \{a, b, a + b\}.$$

Corollary 3.4 (Disjoint additive triples). *If two distinct packages occur in a hypothetical common-root collision, then*

$$\mathcal{T}(a, b) \cap \mathcal{T}(c, d) = \emptyset. \quad (3.3)$$

Thus all six integers

$$a, b, a + b, c, d, c + d$$

are pairwise distinct.

Proof. Choose one collision edge from each package. Every member of an additive triple is either an endpoint or the gap of its selected edge. A shared endpoint contradicts the matching conclusion of Theorem 3.1. A shared gap becomes a shared endpoint after replacing z by $1/z$. For the endpoint-gap case, write the second edge as $\{r, r+s\}$ and suppose, for example, that s is an endpoint of the first edge $\{s, w\}$. Then

$$L_r = L_{r+s}, \quad L_s = L_w,$$

and Lemma 3.3 gives $L_r = L_{r+w}$, producing three distinct equal values. The other cross cases are symmetric. These exhaust the possibilities. $\square$

This simultaneously eliminates equal totals, equal gaps, shared entries, and all endpoint-gap cross equalities.

4 Finite support of every factor orbit and local finiteness

For a primitive irreducible polynomial $h \in \mathbb{Z}[x]$ with $h(0) \neq 0$, define its normalized reciprocal by

$$h^\vee(x) = \text{prim}\left(x^{\deg h} h(1/x)\right),$$

where the sign is chosen so that the leading coefficient is positive. We write $[h] = \{h, h^\vee\}$ for its reciprocal orbit. Since a package product is reciprocal, occurrence of either member implies occurrence of both in the product.

The following elementary estimate is independent of the p -adic theory. It is the strongest global finiteness statement obtained in this round.

Theorem 4.1 (Finite collision support of a fixed point). *Fix $z \in \mathbb{C}^\times$ with $0 < r = |z| < 1$. If $L_m(z) = L_n(z)$, $0 < m < n$, and $k = n - m$, then*

$$\frac{k}{m} \leq \frac{2r^m}{1 - r^m}, \tag{4.1}$$

and

$$n \leq \frac{2m}{1 - r^m}. \tag{4.2}$$

In particular, $\mathcal{G}_z$ has only finitely many edges, and those edges are effectively bounded in terms of r .

Proof. The addition law and the collision $L_m = L_{m+k}$ give $L_m = z^m L_k$. Hence

$$k(1 - z^m) = mz^m(1 - z^k).$$

Taking absolute values yields

$$\frac{k}{m} = r^m \frac{|1 - z^k|}{|1 - z^m|} \leq r^m \frac{1 + r^k}{1 - r^m} \leq \frac{2r^m}{1 - r^m},$$

which is (4.1). Since $k \geq 1$, this forces $1 \leq 2mr^m/(1 - r^m)$; the right side tends to zero, so only finitely many m occur. The original equality also gives

$$n|1 - z^m| = m|1 - z^n| \leq 2m,$$

and (4.2) follows from $|1 - z^m| \geq 1 - r^m$. $\square$

For example, one may take

$$M(r) = \max \left\{ m \geq 1 : \frac{2mr^m}{1-r^m} \geq 1 \right\},$$

with the convention $M(r) = 0$ when the set is empty. Then every edge has $m \leq M(r)$, followed by the explicit bound (4.2).

For a fixed package $\{c, d\}$, define

$$E_{c,d}(X) = \#\{ \{a, b\} : a + b \leq X, \gcd(P_{a,b}, P_{c,d}) \neq 1 \}.$$

Theorem 4.2 (Finite support of factors; local finiteness). *Every noncyclotomic reciprocal irreducible orbit $[h]$ occurs in only finitely many package products $P_{a,b}$. Consequently every package has only finitely many sharing partners. In particular, for each fixed $\{c, d\}$,*

$$E_{c,d}(X) = O_{c,d}(1).$$

Proof. Choose a member of $[h]$ having a root α with $0 < |\alpha| < 1$; this is possible because there are no unit-circle roots, and replacing h by $h^\vee$ inverts all roots. If $[h]$ occurs in $P_{a,b}$, reciprocity of the product ensures that α is a root of $P_{a,b}$. It therefore gives one of the collision edges $\{a, a+b\}$ or $\{b, a+b\}$. Theorem 4.1 leaves only finitely many such edges and hence finitely many unordered packages.

A fixed package product has only finitely many irreducible reciprocal orbits. The union of their finite supports is exactly its set of sharing partners, so that set is finite. $\square$

Corollary 4.3 (Finite decidability of privacy). *For every fixed unordered pair $\{c, d\}$, whether it has a globally private factor is decidable by a finite exact calculation. More precisely, after factoring $P_{c,d}$, choose for each reciprocal factor orbit an algebraic root in the open unit disk and a certified rational upper bound $\rho < 1$ for its modulus. Replacing r by ρ in Theorem 4.1 gives an explicit finite list of packages, and exact polynomial gcds decide all remaining comparisons.*

Remark 4.4. The bounds obtained from a root modulus need not be practical or uniform in c, d . The point is logical: target (a), package by package, is no longer an intrinsically infinite comparison. A uniform proof that at least one orbit survives the finite comparison is still missing.

5 The divided-difference sign law

After replacing z by $1/z$ and replacing every edge by its complementary edge, we may assume $0 < |z| < 1$. Put $R = |z|^2$.

Theorem 5.1 (Sign law). *Suppose $L_m(z) = L_n(z) = A$, where $0 < m < n$ and $0 < |z| < 1$. Then for every real $t \notin \{0, m, n\}$,*

$$\operatorname{sgn}(|1 - tA|^2 - R^t) = \operatorname{sgn}(t(t-m)(t-n)). \quad (5.1)$$

Proof. Let $F_A(t) = |1 - tA|^2 - R^t$. We have $F_A(0) = F_A(m) = F_A(n) = 0$, and

$$F_A'''(t) = -(\log R)^3 R^t > 0.$$

The third divided-difference mean-value theorem gives

$$\frac{F_A(t)}{t(t-m)(t-n)} = F_A[t, 0, m, n] = \frac{F_A'''(\xi)}{3!} > 0$$

for some ξ in the convex hull of the four nodes. $\square$

In particular, the affine interpolation $1 - tA$ lies strictly inside the circle $|w| = |z|^t$ for $m < t < n$, and strictly outside it for $0 < t < m$ and $t > n$.

6 Exact tail conjugacy and the strict chord obstruction

This section contains the main analytic advance of the second attack.

Theorem 6.1 (Exact tail-conjugacy identity). *Assume*

$$L_m(z) = L_{m+a}(z) = A, \quad m, a \geq 1.$$

Then

$$A = z^m L_a(z), \tag{6.1}$$

and, for every integer $j \geq 1$,

$$L_{m+j}(z) - A = \frac{j}{m+j} z^m (L_j(z) - L_a(z)). \tag{6.2}$$

The identity extends to every positive real j after choosing a logarithm of z and defining z^j continuously.

Proof. Apply (3.2) with $(u, v) = (m, a)$:

$$(m + a)L_{m+a} = mL_m + az^m L_a.$$

Since $L_{m+a} = L_m$, this yields (6.1). Applying (3.2) with $(u, v) = (m, j)$, subtracting $(m + j)A$, and using (6.1), gives

$$(m + j)(L_{m+j} - A) = jz^m (L_j - L_a),$$

which is (6.2). □

Thus a loop on the interval $[m, m + a]$ conjugates the entire tail, after an affine translation and a simple scalar weight, to the original sequence centered at its gap a .

Corollary 6.2 (Shifted one-crossing law). *Under the hypotheses of Theorem 6.1, for every real $j > 0$,*

$$\operatorname{sgn}(|1 - jL_a(z)|^2 - |z|^{2j}) = \operatorname{sgn}(j - a). \tag{6.3}$$

Proof. From (6.1),

$$1 - (m + j)A = z^m(1 - jL_a).$$

Therefore

$$F_A(m + j) = |z|^{2m} (|1 - jL_a|^2 - |z|^{2j}).$$

For $j \neq a$, Theorem 5.1, with $n = m + a$, gives (6.3). At $j = a$, both sides are zero directly; here $\operatorname{sgn}(0) = 0$. □

Now suppose a second loop occurs later. Since the four endpoints are distinct, we may choose the smallest endpoint and write the configuration as

$$L_m = L_{m+a}, \quad L_{m+s} = L_{m+t}, \quad 0 < s < t. \tag{6.4}$$

Corollary 6.3 (Strict chord constraint). *Every configuration (6.4) satisfies*

$$\frac{s}{m+s}(L_s - L_a) = \frac{t}{m+t}(L_t - L_a), \quad (6.5)$$

and therefore

$$L_t = (1 - \lambda)L_a + \lambda L_s, \quad \lambda = \frac{s(m+t)}{t(m+s)} \in (0, 1). \quad (6.6)$$

Moreover, $L_s \neq L_a$, so L_t lies in the open line segment joining L_a and L_s .

Proof. Apply (6.2) with $j = s, t$, then use $L_{m+s} = L_{m+t}$. This gives (6.5), and solving it gives (6.6). The inequalities $0 < \lambda < 1$ are equivalent to $0 < s < t$. The four edge endpoints are distinct, so $s \neq a$. If $L_s = L_a$, Lemma 3.3 produces $L_m = L_{m+s}$; together with $L_m = L_{m+a}$ this gives three distinct equal values, contradicting Theorem 3.1. $\square$

There is an equivalent block-centroid form. Put $b = t - s$. The two collisions imply

$$z^s L_b = \frac{m}{m+s} L_a + \frac{s}{m+s} L_s. \quad (6.7)$$

Thus the average of the geometric block of length b beginning at s lies strictly between the two prefix averages L_a and L_s , after the common factor $1 - z$ is removed.

Clearing denominators in (6.5) gives a useful sparse algebraic residue of the two-loop problem:

$$a(m+s)z^t - a(m+t)z^s - m(t-s)z^a + (a+m)(t-s) = 0. \quad (6.8)$$

Any surviving counterexample must satisfy both the original trinomial $H_{m,m+a}(z) = 0$ and the quadrinomial (6.8), as well as the strict segment condition (6.6) and the shifted sign law (6.3).

7 Phase localization, a positive cone, and the real-root subproblem

Write a collision as $L_m = L_{m+a}$, and let $z = re^{i\theta}$, $0 < r < 1$, with $\theta \in (-\pi, \pi]$. The trinomial equation has the exact form

$$z^m(m+a-mz^a) = a. \quad (7.1)$$

The denominator on the right of $z^m = a/(m+a-mz^a)$ lies in a disk centered at the positive real number $m+a$, of radius mr^a .

Theorem 7.1 (Exact phase bounds). *For every collision $L_m = L_{m+a}$ with $0 < |z| = r < 1$,*

$$\text{dist}(m\theta, 2\pi\mathbb{Z}) \leq \arcsin\left(\frac{mr^a}{m+a}\right). \quad (7.2)$$

If a second collision is written as

$$L_{m+s} = L_{m+t}, \quad 0 < s < t,$$

then, with $b = t - s$,

$$\text{dist}((m+s)\theta, 2\pi\mathbb{Z}) \leq \arcsin\left(\frac{(m+s)r^b}{m+t}\right), \quad (7.3)$$

$$\text{dist}(s\theta, 2\pi\mathbb{Z}) \leq \arcsin\left(\frac{mr^a}{m+a}\right) + \arcsin\left(\frac{(m+s)r^b}{m+t}\right). \quad (7.4)$$

If in addition $m < a$, the first bound is strictly smaller than $\pi/6$.

Proof. The largest possible argument of a point in a disk centered at a positive real number R with radius $\rho < R$ is $\arcsin(\rho/R)$. Apply this to (7.1). The second collision gives

$$z^{m+s}(m+t-(m+s)z^{t-s})=t-s,$$

which proves (7.3). Circular distance to $2\pi\mathbb{Z}$ is subadditive, so subtracting the two multiples of θ gives (7.4). If $m < a$, then $mr^a/(m+a) < 1/2$. $\square$

Remark 7.2 (Two useful normalizations cannot be conflated). Replacing z by $1/z$ swaps the lower endpoint and the gap of every edge. Thus one may orient a two-loop configuration so that its smallest entry among the two lowers and two gaps is a lower endpoint. This operation, however, also replaces $|z|$ by $|z|^{-1}$. Without an additional argument one cannot simultaneously impose that global-minimum normalization and $|z| < 1$. The stronger $\pi/6$ estimate is therefore conditional on $m < a$ in the inside orientation, not an automatic global normalization.

The quadrinomial in (6.8) has a useful decomposition. Continue to write $b = t - s$, and let its left side be $R_{m,a;s,t}(x)$.

Proposition 7.3 (Positive-cone decomposition). *One has the exact identity*

$$R_{m,a;s,t}(x) = \frac{a(m+t)}{t}H_{s,t}(x) + \frac{m(t-s)}{t}H_{a,t}(x). \quad (7.5)$$

If $t > a$, then

$$\frac{R_{m,a;s,t}(x)}{(x-1)^2}$$

is a polynomial all of whose coefficients are strictly positive. At a surviving common root,

$$\frac{H_{s,t}(z)}{H_{a,t}(z)} = -\frac{m(t-s)}{a(m+t)} \in \mathbb{R}_{<0}. \quad (7.6)$$

Proof. Expanding the right side of (7.5) gives (6.8). For $0 < u < v$,

$$\frac{H_{u,v}(x)}{(x-1)^2} = \sum_{j=0}^{u-1} (v-u)(j+1)x^j + \sum_{j=u}^{v-2} u(v-j-1)x^j, \quad (7.7)$$

so all coefficients are positive. When $t > a$, both kernels in (7.5) are of this form and both scalar coefficients are positive. At a root of $R_{m,a;s,t}$, neither kernel value can vanish: otherwise both would vanish and $L_a = L_s = L_t$, contrary to the no-three theorem (the indices are distinct in a genuine configuration). Dividing (7.5) therefore gives (7.6). $\square$

Thus the disjoint and interlacing interval patterns reduce to a concrete sector problem: conditional on $H_{m,m+a}(z) = 0$, the two kernel values in (7.6) would have to lie on opposite rays. The nested case $t < a$ remains algebraically different.

There is a further ordering theorem when the common root is real.

Proposition 7.4 (Ordering of two hypothetical negative-real loops). *Let $z = -r$, $0 < r < 1$. Every collision edge $\{m, n\}$ has m even and n odd. If two distinct collision edges satisfy $m < p$, then*

$$m < p, \quad n < q, \quad n - m > q - p. \quad (7.8)$$

In particular, the later edge has larger total and smaller gap; the nested interval pattern is impossible.

Proof. Put

$$E(X) = \frac{1 - e^{-X}}{X}, \quad O(Y) = \frac{1 + e^{-Y}}{Y}.$$

Both functions are strictly decreasing on $(0, \infty)$. Values of $L_k(-r)$ on even and odd indices are respectively $uE(uk)$ and $uO(uk)$, where $u = -\log r$. Equal values cannot have the same parity; an odd lower and an even upper are also impossible because $L_m(-r) > 1/m > 1/n > L_n(-r)$. Hence the lower index is even and the upper one odd.

For each $X > 0$, the equation $E(X) = O(Y)$ has a unique solution $Y > X$. Write it as $Y = \phi(X)$, and let the common value be c . Using

$$E'(X) = \frac{1 - c(X+1)}{X}, \quad O'(Y) = \frac{1 - c(Y+1)}{Y},$$

implicit differentiation gives

$$0 < \phi'(X) = \frac{Y}{X} \frac{c(X+1) - 1}{c(Y+1) - 1} < 1;$$

the last inequality is equivalent to $(1 - c)Y > (1 - c)X$. Thus ϕ is increasing while $\phi(X) - X$ is strictly decreasing. Applying this at $X = um, up$ proves $n < q$ and $n - m > q - p$. $\square$

This does not yet exclude two negative-real loops, but it reduces them to a strictly ordered tradeoff that is invisible in the general complex case.

8 Schur, Casoratian, and contraction formulations

Define the bivariate deformation

$$\begin{aligned} \mathcal{F}_{m,n}(x, t) &= (1 - t^m)x^n + (t^n - 1)x^m + t^m - t^n \\ &= \det \begin{pmatrix} 1 & t^n & x^n \\ 1 & t^m & x^m \\ 1 & 1 & 1 \end{pmatrix}. \end{aligned} \tag{8.1}$$

It is divisible by $(t - 1)(x - 1)(x - t)$. Put

$$\mathcal{R}_{m,n}(x, t) = \frac{\mathcal{F}_{m,n}(x, t)}{(t - 1)(x - 1)(x - t)}.$$

The alternant formula for Schur polynomials gives

$$\mathcal{R}_{m,n}(x, t) = -s_{(n-2, m-1)}(1, t, x). \tag{8.2}$$

At $t = 1$,

$$\mathcal{R}_{m,n}(x, 1) = -\frac{H_{m,n}(x)}{(x - 1)^2}. \tag{8.3}$$

If $e = \gcd(m, n)$ and $[e](x) = (x^e - 1)/(x - 1)$, the exact relation to our polynomial is

$$\mathcal{R}_{m,n}(x, 1) = -Q_{m, n-m}(x)[e](x)^2. \tag{8.4}$$

Thus the specialization is literally $-Q$ in the primitive case and differs from it only by an explicit cyclotomic square in general.

For the one-variable specialization put

$$u_k(x) = h_k(1, 1, x) = \sum_{j=0}^k (k-j+1)x^j = \frac{x^{k+2} - (k+2)x + k+1}{(x-1)^2}, \quad (8.5)$$

with $u_{-1} = 0$. Jacobi–Trudi gives

$$\frac{H_{m,n}(x)}{(x-1)^2} = u_{n-2}u_{m-1} - u_{n-1}u_{m-2}. \quad (8.6)$$

The sequence satisfies

$$u_{k+1} = xu_k + k + 2 \quad (k \geq -1), \quad u_k - 2u_{k-1} + u_{k-2} = x^k \quad (k \geq 1). \quad (8.7)$$

If $v_k = (u_k, u_{k+1})$, then

$$\det(v_i, v_j) = (j+2)u_i - (i+2)u_j. \quad (8.8)$$

Hence a collision is a repeated projective value in the sequence $[v_k]$.

Equivalently, let

$$r_k(x) = \frac{u_{k-2}(x)}{k}, \quad r_1 = 0.$$

Then

$$r_{k+1} = \frac{k}{k+1}(1 + xr_k), \quad (8.9)$$

and

$$x \neq 1 : \quad H_{m,n}(x) = 0 \iff r_m(x) = r_n(x). \quad (8.10)$$

Indeed,

$$r_k(x) = -\frac{L_k(x)}{(x-1)^2} - \frac{1}{x-1}. \quad (8.11)$$

For $|x| < 1$, every map

$$T_k(w) = \frac{k}{k+1}(1 + xw)$$

is a strict affine contraction. A collision says that the orbit beginning at $r_1 = 0$ hits the unique fixed point of an interval composition. Formula (6.2) is the exact affine conjugacy of the tail induced by such a hit.

Plücker relations by themselves only turn two vanishing minors into a product identity. The recurrences (8.7) do not yet yield the desired descent to a third vanishing minor. The strict chord identity (6.6) is the strongest closed consequence obtained from this route in the present round.

9 The correct second jet of the Schur deformation

A first-order transversality argument cannot work: all reduced Schur curves have the same tangent at $t = 1$. The correct coordinates are

$$t = e^\tau, \quad x = e^{\tau/2}y.$$

If $D = |\lambda|$ denotes the size of the partition, homogeneity and symmetry in the first two variables give

$$e^{-D\tau/2}s_\lambda(1, e^\tau, e^{\tau/2}y) = s_\lambda(e^{-\tau/2}, e^{\tau/2}, y),$$

and the right side is even in τ .

For our two-row partition define

$$S_{m,n}(y, \tau) = s_{(n-2, m-1)}(e^{-\tau/2}, e^{\tau/2}, y).$$

The bialternant formula yields

$$S_{m,n}(y, \tau) = \frac{\sinh(m\tau/2)y^n - \sinh(n\tau/2)y^m + \sinh((n-m)\tau/2)}{\sinh(\tau/2)(y^2 - 2\cosh(\tau/2)y + 1)}. \quad (9.1)$$

At $\tau = 0$, this is $H_{m,n}(y)/(y-1)^2$.

Proposition 9.1 (Second-jet root branch). *Suppose z is a root of $H_{m,n}$ with $|z| \neq 1$, and put*

$$A = L_m(z) = L_n(z).$$

Then the local root branch of $S_{m,n}(y, \tau) = 0$ through $(z, 0)$ has the form

$$y_{m,n}(\tau) = z + \frac{z}{24} \left(m + n - \frac{3}{A} \right) \tau^2 + O(\tau^4). \quad (9.2)$$

Proof. Expanding (9.1) at $\tau = 0$ and using $H_{m,n}(z) = 0$ gives

$$S_{m,n}(z, \tau) = \frac{\tau^2}{24(z-1)^2} (m^3 z^n - n^3 z^m + (n-m)^3) + O(\tau^4). \quad (9.3)$$

Since $z^m = 1 - mA$ and $z^n = 1 - nA$, the bracket equals

$$mn(n-m)((m+n)A - 3). \quad (9.4)$$

On the other hand,

$$\begin{aligned} \frac{\partial}{\partial y} \frac{H_{m,n}(y)}{(y-1)^2} \Big|_{y=z} &= \frac{mnz^{m-1}(z^{n-m} - 1)}{(z-1)^2} \\ &= -\frac{mn(n-m)A}{z(z-1)^2}. \end{aligned} \quad (9.5)$$

Here $A \neq 0$, because $A = 0$ would give $z^m = 1$, contrary to $|z| \neq 1$. Hence the derivative is nonzero. The implicit-function theorem gives a unique local root branch. Since $S_{m,n}(y, \tau) = S_{m,n}(y, -\tau)$, uniqueness forces $y_{m,n}(\tau) = y_{m,n}(-\tau)$; the branch is even. Combining this with (9.3) – (9.5) gives (9.2) with remainder $O(\tau^4)$. $\square$

If two edges $\{m, n\}$, $\{p, q\}$ share a root z , with collision values A, B , their comoving branches therefore separate as

$$y_{m,n}(\tau) - y_{p,q}(\tau) = \frac{z}{24} \left(m + n - p - q - 3 \left(\frac{1}{A} - \frac{1}{B} \right) \right) \tau^2 + O(\tau^4). \quad (9.6)$$

In the comoving local parameter τ , the two root branches therefore differ by $O(\tau^2)$, and the order is exactly two unless the explicit bracket in (9.6) vanishes. For example, if $m + n = p + q$, higher contact would force $A = B$, contradicting the no-three theorem for disjoint edges. This is the correct second-order replacement for the incorrect first-order “transversality” formulation.

10 A complete classification of rational roots

We first record the coefficient expansion, valid when $\gcd(a, b) = 1$:

$$Q_{a,b}(x) = b \sum_{j=0}^{a-1} (j+1)x^j + a \sum_{j=a}^{a+b-2} (a+b-1-j)x^j. \quad (10.1)$$

In particular, the constant term is b , the leading coefficient is a , and all coefficients are positive.

Theorem 10.1 (Rational-root classification). *For arbitrary positive a, b , the polynomial $Q_{a,b}$ has a rational root if and only if*

$$(a, b) = (2, 1) \quad \text{or} \quad (a, b) = (1, 2).$$

The roots are respectively $-1/2$ and -2 .

Proof. First suppose $\gcd(a, b) = 1$. By positivity of the coefficients there is no positive root. By reciprocity it suffices to classify a rational root $x = -r/s \in (-1, 0)$, with r, s coprime positive integers and $r < s$. Write $y = r/s \in (0, 1)$ and $N = a + b$. If a is odd and N is even, then

$$H_{a,N}(-y) = ay^N + Ny^a + b > 0.$$

If a and N are both odd, then

$$H_{a,N}(-y) = -ay^N + Ny^a + b > 0,$$

because $y^a > y^N$ and $N > a$. If a, N are both even, then

$$H_{a,N}(-y) = b - Ny^a + ay^N > 0;$$

its derivative is $aNy^{a-1}(y^{N-a} - 1) < 0$, and it decreases from b to 0. Hence only a even and N odd can remain; in particular b is odd.

The rational-root theorem applied to (10.1) gives

$$r \mid b, \quad s \mid a.$$

Writing $N = a + b$, the equation $H_{a,N}(-r/s) = 0$ becomes

$$bs^{a+b} = (a+b)r^a s^b + ar^{a+b}. \quad (10.2)$$

Let $\ell \mid s$, and put $t = v_\ell(s)$. Since $\ell \nmid b(a+b)r$, the two terms on the right of (10.2) have valuations bt and $v_\ell(a)$, whereas the left side has valuation $(a+b)t > bt$. If the two right-hand valuations were unequal, the valuation of their sum would be their smaller value, strictly below that of the left side. They must therefore coincide, giving

$$v_\ell(a) = b v_\ell(s). \quad (10.3)$$

Likewise, if $\ell \mid r$ and $t = v_\ell(r)$, then the two right-hand valuations are at and $(a+b)t$, while the left side has valuation $v_\ell(b)$. Hence

$$v_\ell(b) = a v_\ell(r). \quad (10.4)$$

Consequently,

$$s^b \mid a, \quad r^a \mid b. \quad (10.5)$$

If $r \geq 2$, then $b \geq 2^a > a$, while $s \geq 2$ and (10.5) give $a \geq 2^b > b$, a contradiction. Hence $r = 1$.

Write $a = s^b A$ with $\gcd(A, s) = 1$. Dividing (10.2) by s^b gives

$$b(s^a - 1) = A(s^b + 1) = a + A \leq \frac{3a}{2}. \quad (10.6)$$

For $a \geq 3$, the left side is at least $2^a - 1 > 3a/2$. Therefore $a = 2$, and (10.5) gives $s = 2$, $b = 1$, $A = 1$. Thus the only primitive root in the open unit disk is $-1/2$ for $Q_{2,1}$. Reciprocity gives -2 for $Q_{1,2}$.

Finally, for a nonprimitive pair write $a = gA$, $b = gB$. By (2.5), a rational root x would make x^g a rational root of a primitive $Q_{A,B}$. The only possibilities are $-1/2$ and -2 . If $g > 1$, then g must be odd, and the equality $g v_2(x) = \pm 1$ is impossible for rational x . Hence $g = 1$, and the two cases above are the only cases. $\square$

This theorem closes the degree-one exception in the p -adic argument below.

11 Full additive-triple prime separation

We first isolate the two standard local-field facts used below; see, for example, [8, Chapters I–II].

Lemma 11.1 (Two local-field criteria). *Let p be a prime.*

- (a) *A root of unity of order prime to p belongs to $\mathbb{Q}_p^{\text{nr}}$. If ζ is such a root of unity and $v_p(\beta - \zeta) \notin \mathbb{Z}$, then $\mathbb{Q}_p(\beta)/\mathbb{Q}_p$ is ramified.*
- (b) *A monic polynomial in $\mathbb{Z}_p[x]$ with unit constant term and square-free reduction has only unit roots, and every root lies in a finite unramified extension of $\mathbb{Q}_p$.*

Proof. For (a), prime-to- p roots of unity are Teichmüller lifts and lie in the maximal unramified extension. If β were also contained in an unramified extension, then the compositum containing β and ζ would be unramified, whose normalized value group is $\mathbb{Z}$, contradicting the nonintegral valuation of their difference. Part (b) is the usual Hensel lifting description of finite étale $\mathbb{Z}_p$ -algebras. Monicity makes every root integral, and the unit constant term forces all root valuations to be zero. $\square$

Lemma 11.2 (Good package away from its additive triple). *Let p be odd. If $p \nmid cd(c + d)$, then every root of both orientations in $\mathcal{P}_{\{c,d\}}$ is a p -adic unit lying in a finite unramified extension of $\mathbb{Q}_p$.*

Proof. Write $c = hC$, $d = hD$, with $\gcd(C, D) = 1$. Then $p \nmid hCD(C + D)$. Borisov's discriminant formula [1, Lemma 2.1] shows that the primitive polynomial $Q_{C,D}$, after division by its unit leading coefficient, is monic over $\mathbb{Z}_p$, has unit constant term, and has square-free reduction. Furthermore,

$$\frac{d}{dx} Q_{C,D}(x^h) = hx^{h-1} Q'_{C,D}(x^h).$$

At a root modulo p , one has $x \neq 0$, $p \nmid h$, and the last factor is nonzero because the primitive reduction is square-free. Thus $Q_{C,D}(x^h)$ also has square-free reduction. Equation (2.5) differs from this composition only by a scalar. Lemma 11.1(b) applies, and the reciprocal orientation is identical. $\square$

For reference, Borisov's Lemmas 2.8–2.11 give the following consequence for the primitive polynomial $Q_{A,B} = f_{A+B,A,B}$.

Lemma 11.3 (Ramification supplied by a primitive additive-triple prime). *Let $\gcd(A, B) = 1$, let $N = A + B$, and suppose that $p \mid ABN$.*

- (a) *If $p \geq 5$, every root of either orientation is either a nonunit or lies at nonintegral-valued distance from a root of unity of order prime to p . In the latter case its local field is ramified.*
- (b) *If $p = 3$, each fixed orientation has at most one unit root in $\mathbb{Q}_3^{\text{nr}}$; every other root is either a nonunit or ramified.*

Proof. The three numbers A, B, N are pairwise coprime, so p divides exactly one of them. If $p \mid N$, Lemmas 2.8–2.9 of [1] place all roots in clusters around prime-to- p roots of unity. At nontrivial centers the successive distances have valuations

$$\frac{1}{p}, \frac{1}{p^2 - p}, \dots,$$

and at the center 1 the first layer has valuation $1/(p - 2)$, followed by the same higher layers. For $p \geq 5$ all are nonintegral. For $p = 3$, the first layer at 1 consists of exactly one root at integral valuation; all other layers are fractional.

If $p \mid B$, Borisov's Lemma 2.10 gives A nonunit roots and places the remaining roots in the same root-of-unity clusters. If $p \mid A$, Lemma 2.11 gives the reciprocal statement. Lemma 11.1(a) converts every fractional cluster distance into ramification. The same analysis applies after interchanging the two orientations. $\square$

Theorem 11.4 (Full additive-triple prime separation). *Write $a = gA, b = gB$, with $\gcd(A, B) = 1$. Suppose*

$$p \mid AB(A + B), \quad p \nmid cd(c + d). \quad (11.1)$$

If $p \geq 5$, then

$$\gcd_{\mathbb{Q}[x]}(P_{a,b}, P_{c,d}) = 1. \quad (11.2)$$

If $g = 1$, the same conclusion holds for $p = 3$.

Proof. Suppose an irreducible $h \in \mathbb{Q}[x]$ divides both package products. Fix an orientation of each package divisible by h , and choose a root $\alpha \in \overline{\mathbb{Q}}_p$ of h . Lemma 11.2 shows that α is a unit in an unramified extension. Put $\beta = \alpha^g$. Then β is a root of the corresponding primitive orientation $Q_{A,B}$ or $Q_{B,A}$, and it is again a unit in an unramified extension. For $p \geq 5$, this contradicts Lemma 11.3(a).

Now let $p = 3$ and $g = 1$. Every root of h is a root of the good second orientation and therefore lies in $\mathbb{Q}_3^{\text{nr}}$. The fixed first orientation has at most one such unit root by Lemma 11.3(b). Since h is separable, $\deg h \leq 1$. The rational-root classification (Theorem 10.1) forces the package $\{1, 2\}$. The same theorem would force the second package to be $\{1, 2\}$ as well, contrary to $3 \nmid cd(c + d)$. $\square$

Corollary 11.5 (Support compatibility). *Fix $\{c, d\}$ and let*

$$S = \{p : p \mid 6cd(c + d)\}.$$

If $P_{a,b}$ shares a factor with $P_{c,d}$, and $a = gA, b = gB$ is its primitive reduction, then

$$\text{Supp}(AB(A + B)) \subseteq S. \quad (11.3)$$

If both packages are primitive, their primitive additive-triple products have the same prime support outside $\{2, 3\}$.

Proof. Every prime $p \geq 5$ in the left side but outside S is excluded by Theorem 11.4; the primes 2, 3 were inserted into S . For two primitive packages, apply the theorem in both directions. $\square$

12 The S -unit reduction and finite primitive shapes

The support statement is much stronger than smoothness of the primitive total: all three pairwise coprime entries $A, B, A + B$ are supported on one fixed finite set.

Theorem 12.1 (Finite primitive-shape theorem). *Fix $\{c, d\}$ and let S be as in Corollary 11.5. There are only finitely many ordered primitive pairs (A, B) that can occur as primitive reductions of sharing partners. If $s = |S|$, their number is at most*

$$2^{16s+8}. \quad (12.1)$$

Proof. Let $N = A + B$, and put

$$u = \frac{A}{N}, \quad v = \frac{B}{N}.$$

By (11.3), both u and v belong to the rational S -unit group U_S , and $u + v = 1$. The group $U_S \times U_S$ has rank $2s$. The theorem of Beukers–Schlickewei [3] bounds the number of solutions of $x + y = 1$ in a multiplicative subgroup of $(\mathbb{C}^\times)^2$ of rank r by 2^{8r+8} , giving (12.1). The map $(A, B) \mapsto (A/N, B/N)$ is injective on primitive ordered pairs, because each fraction is already in lowest terms. $\square$

There is an elementary final step from shapes to actual scaled packages.

Proposition 12.2 (Finite scaling for a fixed shape). *Fix a package $P_{c,d}$ and a primitive shape (A, B) . Only finitely many positive integers g can make $P_{gA,gB}$ share a factor with $P_{c,d}$.*

Proof. A shared factor supplies a root α of $P_{c,d}$ such that $\beta = \alpha^g$ is a root of $P_{A,B}$. There are finitely many choices of (α, β) , and $|\alpha| \neq 1$. For a fixed pair, $|\alpha|^g = |\beta|$ determines at most one positive integer g . $\square$

Theorem 12.1 and Proposition 12.2 give a second proof of local finiteness. Unlike the direct Archimedean proof, they also describe the arithmetic shape of the finite neighborhood. This is the main reason target (a) deserves renewed attention: failure of privacy for $P_{c,d}$ is now a finite divisibility assertion among explicitly constrained S -unit shapes and finitely many scalings.

13 Global proper-divisibility rigidity

The next theorem now includes the boundary primitive shapes. Its principal algebraic input is an exact denominator calculation.

Lemma 13.1 (Exact diagonal denominators). *Let a, b be coprime positive integers, put $N = a + b$, and set*

$$F(x) = 1 - \frac{N}{b}x^a + \frac{a}{b}x^N.$$

For every $k \geq 0$,

$$[x^{ka}]F(x)^{-1} = \frac{C_k}{b^k}, \quad (13.1)$$

where

$$C_k = N^k + \sum_{1 \leq r \leq \lfloor k/N \rfloor} \binom{k-br}{ar} N^{k-Nr} (-a)^{ar} b^{br}. \quad (13.2)$$

In particular, $\gcd(C_k, b) = 1$, so the coefficient in (13.1) has denominator exactly b^k .

Proof. Expand geometrically:

$$F(x)^{-1} = \sum_{j \geq 0} \left(\frac{N}{b} x^a - \frac{a}{b} x^N \right)^j.$$

If exactly t of the j selected monomials have exponent N , the resulting exponent is $(j-t)a + tN = ja + tb$. For this to equal ka , coprimality forces

$$t = ar, \quad j = k - br$$

for an integer $r \geq 0$, and $t \leq j$ is equivalent to $Nr \leq k$. The corresponding contribution is

$$\binom{k-br}{ar} \frac{N^{k-Nr} (-a)^{ar}}{b^{k-br}}.$$

Putting all terms over b^k gives (13.2). Every term with $r \geq 1$ is divisible by every prime dividing b , whereas $N^k \equiv a^k \not\equiv 0 \pmod{p}$ for $p \mid b$. Thus C_k is coprime to b . $\square$

Theorem 13.2 (Proper-divisibility rigidity). *Let a, b be coprime positive integers with $a \neq b$. For arbitrary positive c, d ,*

$$Q_{a,b}(x) \mid Q_{c,d}(x) \quad \text{in } \mathbb{Q}[x] \quad \implies \quad (c, d) = (a, b). \quad (13.3)$$

Proof. If $c = d$, then $Q_{c,c} = c$ is constant and cannot be divisible by the nonconstant polynomial $Q_{a,b}$. If the unordered packages agree, the no-three theorem applied to the two complementary edges shows that the two orientations are coprime; hence the orientations in (13.3) must agree. We therefore assume that the packages are distinct, so Corollary 3.4 gives disjoint additive triples.

First suppose $a, b > 1$. The polynomial $Q_{a,b}$ is primitive in $\mathbb{Z}[x]$, because its leading and constant coefficients are the coprime integers a, b . Gauss's lemma gives

$$Q_{c,d} = Q_{a,b} R, \quad R \in \mathbb{Z}[x].$$

Comparison of leading and constant coefficients gives

$$c = au, \quad d = bv, \quad u, v \in \mathbb{Z}_{>0}. \quad (13.4)$$

Disjointness of the additive triples gives $u, v \geq 2$.

Let $h = \gcd(c, d)$, let $[h](x) = (x^h - 1)/(x - 1)$, and put $S(x) = [h](x)^2 R(x) \in \mathbb{Z}[x]$. Writing $N = a + b$, the identity between the trinomial numerators is

$$H_{c,c+d}(x) = H_{a,N}(x) S(x).$$

Reduction modulo the ideal $(x^c) = (x^{au})$ gives

$$bv \equiv b \left(1 - \frac{N}{b}x^a + \frac{a}{b}x^N \right) S(x) \pmod{x^{au}},$$

and hence

$$S(x) \equiv v \left(1 - \frac{N}{b}x^a + \frac{a}{b}x^N \right)^{-1} \pmod{x^{au}}. \quad (13.5)$$

For $0 \leq k < u$, the coefficient of x^{ka} on the left is integral. Lemma 13.1 therefore gives $b^k \mid v$, and at $k = u - 1$,

$$b^{u-1} \mid v. \quad (13.6)$$

Reciprocating the divisibility and repeating the argument gives

$$a^{v-1} \mid u. \quad (13.7)$$

Thus $v \geq 2^{u-1}$ and $u \geq 2^{v-1}$. If, say, $u \leq v$, then $u \geq 2^{v-1} \geq 2^{u-1}$; for $u \geq 3$ this is impossible. Hence $u = 2$, and then $2 \geq 2^{v-1}$ gives $v = 2$. Thus $u = v = 2$. The assumed divisibility then gives $Q_{a,b}(x) \mid Q_{a,b}(x^2)$. Squaring maps the finite root set into itself, so every root is eventually periodic under squaring and hence a root of unity, contradicting Lemma 2.2. This eliminates the interior case.

It remains to treat a boundary shape. By reciprocity it is enough to assume $b = 1$ and $a > 1$. Comparison of leading coefficients gives

$$c = au \quad (u \in \mathbb{Z}_{>0}). \quad (13.8)$$

Since the packages are distinct, additive-triple separation gives $d \geq 2$. Reciprocating yields

$$Q_{1,a} \mid Q_{d,c}.$$

Apply Lemma 13.1 to the source parameters $(1, a)$ and target parameters (d, c) . The target leading multiplier is d , and its constant multiplier is $u = c/a$; taking the diagonal coefficient with $k = d - 1$ gives

$$a^{d-1} \mid u, \quad \text{hence} \quad c = au \geq a^d. \quad (13.9)$$

Now

$$Q_{a,1}(x) = 1 + 2x + \cdots + ax^{a-1}.$$

By the Eneström–Kakeya coefficient-ratio bound [4], every root α satisfies

$$r := |\alpha| \leq \frac{a-1}{a} \leq e^{-1/a} < 1. \quad (13.10)$$

The same α is a root of $Q_{c,d}$, so the collision estimate (4.1), applied to the edge $\{c, c+d\}$, gives

$$\frac{d}{c} \leq \frac{2r^c}{1-r^c}, \quad d \leq \frac{2c}{e^{c/a} - 1}. \quad (13.11)$$

The function $x/(e^{x/a} - 1)$ is decreasing for $x > 0$: with $y = x/a$, the numerator of its derivative is $e^y(1-y) - 1 < 0$. Using (13.9) in (13.11), and then $e^y - 1 > y^2$ for $y \geq 2$, yields

$$d \leq \frac{2a^d}{e^{a^{d-1}} - 1} < 2a^{2-d}.$$

For $d = 2$ the right side is strictly smaller than 2, and for $d \geq 3$ it is at most 1, a contradiction. Thus no distinct target exists. Reciprocity treats the case $a = 1$. $\square$

Corollary 13.3 (Primitive irreducibility gives global isolation). *Let a, b be coprime and unequal. If $Q_{a,b}$ is irreducible over $\mathbb{Q}$, then the package $\{a, b\}$ is globally isolated from the entire primitive and nonprimitive family.*

Proof. The reciprocal $Q_{b,a}$ is irreducible as well. A common package factor would make one of these two polynomials divide one orientation of the other package, contradicting Theorem 13.2 unless the packages agree. $\square$

14 The Schinzel fibre bound, self-power rigidity, and isolation

The proper-divisibility theorem controls a whole primitive polynomial. We next make Schinzel's trinomial-fibre argument explicit, then combine it with a local Newton-edge obstruction and a Frobenius-quotient argument.

Lemma 14.1 (Elementary binomial-fibre lemma). *Let $0 < m < n$, let $d \geq 1$, and assume $\gcd(\gcd(m, n), d) = 1$. For every $c \in \mathbb{C}^\times$,*

$$\deg \gcd(H_{m,n}(x), x^d - c) \leq 1.$$

Proof. Suppose that the gcd has degree at least two. Since $x^d - c$ is separable, there are two distinct common roots α and $\zeta\alpha$, where $\zeta^d = 1$ and $\zeta \neq 1$. Put $g = \gcd(m, n)$. The hypothesis gives $\gcd(g, d) = 1$, so $\xi \mapsto \xi^g$ is an automorphism of μ_d . Because

$$H_{m,n}(x) = gH_{m/g, n/g}(x^g),$$

the two transformed roots remain distinct, and it is enough to treat $\gcd(m, n) = 1$. Set $r = n - m$; then m, r, n are pairwise coprime.

Write

$$u = \zeta^m, \quad v = \zeta^n, \quad X = \alpha^m, \quad Y = \alpha^n.$$

The two root equations are

$$mY - nX = -r, \quad mvY - nuX = -r.$$

If $u = v$, these equations give $u = v = 1$, and the coprimality of m, n gives $\zeta = 1$, a contradiction. Hence

$$X = \frac{r(v-1)}{n(v-u)}, \quad Y = \frac{r(u-1)}{m(v-u)}.$$

Since $X^n = Y^m$, one obtains the master identity

$$m^m r^r (\zeta^n - 1)^n = n^n \zeta^{mr} (\zeta^m - 1)^m (\zeta^r - 1)^r. \quad (14.F1)$$

None of $\zeta^m, \zeta^n, \zeta^r$ is 1, by the displayed formulas and $\alpha \neq 0$.

Let $D = \text{ord}(\zeta) = \ell^a s$, where $(\ell, s) = 1$. For each rational prime ℓ , choose a place of $\mathbb{Q}(\zeta)$ above ℓ and normalize its valuation by $v_\ell(\ell) = 1$. The standard cyclotomic valuation formula is

$$v_\ell(1 - \zeta^k) = \begin{cases} 0, & s \nmid k, \\ \frac{1}{\varphi(\ell^{a - \min(a, v_\ell(k))})}, & s \mid k, \zeta^k \neq 1. \end{cases}$$

It follows by separating the prime-to- ℓ Teichmuller part from the ℓ -power part and using $\Phi_{\ell^j}(1) = \ell$. Assume first that $\ell \mid m$, and put $t = v_\ell(m)$. Taking valuations in (14.F1) gives

$$mt + nv_\ell(1 - \zeta^n) = mv_\ell(1 - \zeta^m) + rv_\ell(1 - \zeta^r). \quad (14.F2)$$

If $s = 1$, then $a > t$, because $\zeta^m \neq 1$, and the valuation identity becomes

$$t + \frac{1}{\varphi(\ell^a)} = \frac{1}{\varphi(\ell^{a-t})},$$

or

$$t\varphi(\ell^a) + 1 = \ell^t.$$

This is impossible: since $a > t$, $t\ell^{a-1}(\ell - 1) + 1 > \ell^t$.

Suppose $s > 1$. Since m, r, n are pairwise coprime, s divides at most one of them. If it divides none, the valuation identity gives $mt = 0$. If $s \mid n$, its left side has an additional positive term and its right side has none. If $s \mid r$, then $mt = r/\varphi(\ell^a)$, so $m \mid r$, contrary to $(m, r) = 1$. Finally, if $s \mid m$, then

$$t = \frac{1}{\varphi(\ell^{a-t})},$$

so necessarily

$$\ell = 2, \quad t = 1, \quad a = 2.$$

If instead $\ell \mid n$, with $t = v_\ell(n)$, valuation of (14.F1) gives

$$nv_\ell(1 - \zeta^n) = nt + mv_\ell(1 - \zeta^m) + rv_\ell(1 - \zeta^r).$$

When $s = 1$, division by $n = m + r$ gives exactly the same impossible identity

$$t + \frac{1}{\varphi(\ell^a)} = \frac{1}{\varphi(\ell^{a-t})}.$$

When $s > 1$, the prime-to- ℓ part s divides at most one of m, r, n . If $s \nmid n$, the right side is strictly positive while the left side is zero; if $s \mid n$, the identity reduces to $t = 1/\varphi(\ell^{a-t})$. Thus again $\ell = 2$, $t = 1$, and $a = 2$.

Finally, if $\ell \mid r$, apply the already proved $\ell \mid m$ case to the reciprocal edge (r, n) : the identity

$$x^n H_{r,n}(x^{-1}) = H_{m,n}(x)$$

sends the two roots $\alpha, \zeta\alpha$ to $\alpha^{-1}, \zeta^{-1}\alpha^{-1}$, and ζ^{-1} has the same order as ζ . Hence every prime divisor of mnr would have to be 2, occurring to the first power. Pairwise coprimality and $n = m + r$ force $(m, r, n) = (1, 1, 2)$. But $H_{1,2}(x) = (x - 1)^2$, which cannot share two distinct roots with the separable polynomial $x^d - c$. This contradiction proves the lemma. $\square$

Theorem 14.2 (Confluent Schinzel degree bound). *Let $0 < m < n$, $0 < p < q$, and put*

$$d_1 = \gcd(m, n), \quad d_2 = \gcd(p, q).$$

If $\gcd(d_1, d_2) = 1$, then

$$\deg \gcd(H_{m,n}, H_{p,q}) \leq \min \left\{ \frac{n}{d_1}, \frac{q}{d_2} \right\}. \quad (14.1)$$

Consequently, if a noncyclotomic irreducible polynomial h of degree e divides both trinomials, then

$$e \leq \min \left\{ \frac{n}{d_1}, \frac{q}{d_2} \right\} - 2. \quad (14.2)$$

Proof. Write $p = d_2 p'$, $q = d_2 q'$. As a polynomial in $y = x^{d_2}$, the second trinomial has degree q' , so over $\mathbb{C}$

$$H_{p,q}(x) = \lambda \prod_{j=1}^{q'} (x^{d_2} - c_j),$$

where the $c_j \neq 0$ are counted with multiplicity. Since $\gcd(d_1, d_2) = 1$, Lemma 14.1 applies to every fibre, and hence

$$\deg \gcd(H_{m,n}, H_{p,q}) \leq \sum_{j=1}^{q'} \deg \gcd(H_{m,n}, x^{d_2} - c_j) \leq q' = \frac{q}{d_2}.$$

Interchanging the two trinomials gives the other half of (14.1). Their only common roots of unity are at 1, with multiplicity two in each. Thus a common noncyclotomic factor of degree e , together with $(x-1)^2$, lies in the gcd, proving (14.2). The only external input in this proof has therefore been removed: the fibre estimate is the elementary Lemma 14.1. $\square$

14.1 Two internal-power obstructions

The first lemma isolates the exact local mechanism used at primes dividing an additive triple.

Lemma 14.3 (Newton-edge power obstruction). *Let $L/\mathbb{Q}_p$ be unramified, let $\zeta \in L$ be a root of unity of order prime to p , and let β be algebraic over L . Suppose that for some integer $e \geq 1$,*

$$v_p(\beta - \zeta) = \frac{1}{e}, \quad [L(\beta) : L] \leq e.$$

Then $\beta \notin L(\beta)^p$.

Proof. The value $1/e$ in the value group forces the ramification index of $L(\beta)/L$ to be divisible by e . The degree hypothesis therefore forces equality throughout: the extension has degree and ramification index exactly e , and its value group is $e^{-1}\mathbb{Z}$.

Suppose $\beta = w^p$ with $w \in L(\beta)$. Frobenius is an automorphism of the residue field of L , so there is a Teichmüller root $\eta \in L$ with $\eta^p = \zeta$. Reduction modulo the maximal ideal gives $w \equiv \eta$. Write $w = \eta + \delta$; then $v_p(\delta) \geq 1/e$. In

$$\beta - \zeta = (\eta + \delta)^p - \eta^p = \sum_{j=1}^{p-1} \binom{p}{j} \eta^{p-j} \delta^j + \delta^p,$$

every intermediate term has valuation at least $1 + j/e > 1/e$, and the last has valuation at least $p/e > 1/e$. This contradicts $v_p(\beta - \zeta) = 1/e$. $\square$

The odd-prime arguments below need only the following root packet. It is best taken directly from Borisov's complete Newton-polygon description rather than inferred from a partial coefficient calculation.

Lemma 14.4 (The center-1 packet in every odd-prime role). *Let A, B be coprime, put $N = A + B$, and let p be odd. If p^κ exactly divides one of A, B, N , then exactly $p - 2$ roots γ of $Q_{A,B}$ satisfy*

$$v_p(\gamma - 1) = \frac{1}{p-2}.$$

Consequently the Newton polygon of $Q_{A,B}(1+Y)$ has a side of horizontal length $p-2$ and slope $-1/(p-2)$.

Proof. Under the dictionary

$$Q_{A,B} = f_{N,A,B}$$

for Borisov's notation $(a, b, c) = (N, A, B)$. If $p \mid N$, the asserted packet is exactly Lemma 2.9 of [1]. If $p \mid B$, Lemma 2.10 first separates the A nonunit roots and states that the remaining roots occur in root-of-unity clusters exactly as in the total role; the cluster centered at 1 therefore contains the same $p - 2$ roots. If $p \mid A$, Lemma 2.11 gives the reciprocal endpoint statement and the same center-1 packet. The final assertion follows from the standard Newton-polygon root-slope correspondence: roots with $v_p(\gamma - 1) = 1/(p - 2)$ contribute a side of slope $-1/(p - 2)$, and their number is its horizontal length. $\square$

For an integer t and a prime p , put

$$q_p(t) = \frac{t^p - t}{p}, \quad C_p(X, Y) = \frac{(X + Y)^p - X^p - Y^p}{p}.$$

Both are integral. The following factor-level theorem is stronger than what is needed for an irreducible whole polynomial.

Theorem 14.5 (Good-prime Frobenius bound for a proper factor). *Let A, B be coprime positive integers, put $N = A + B$, and let h be an irreducible factor of $F = Q_{A,B}$ of degree e . Let β be a root of h . If p is an odd prime with*

$$p \nmid ABN$$

and $\beta \in \mathbb{Q}(\beta)^p$, then

$$e \leq p - 2. \tag{14.3}$$

Proof. Choose $h \in \mathbb{Z}[x]$ primitive with positive leading coefficient and write $F = hk$ in $\mathbb{Z}[x]$. The leading and constant coefficients of h are coprime, because they divide the coprime integers A and B . Suppose $\beta = w^p$ in $K = \mathbb{Q}(\beta)$. Then $K = \mathbb{Q}(w)$, and the norm identity gives

$$\text{lc}(h) = u^p, \quad h(0) = v_0^p$$

for coprime integers $u > 0$ and v_0 , where the sign is absorbed into v_0 because p is odd.

Let $G \in \mathbb{Z}[x]$ be the primitive minimal polynomial of w , with positive leading coefficient t , and put $s = G(0)$. Since $G \mid h(x^p)$, one has $t \mid u^p$ and $s \mid v_0^p$, hence $(t, s) = 1$. The norm identity gives

$$\left(\frac{s}{t}\right)^p = \left(\frac{v_0}{u}\right)^p.$$

The odd p -th power map is injective on $\mathbb{Q}$, and both fractions are reduced. Therefore

$$t = u, \quad s = v_0.$$

In particular, $p \nmid u$, because $u^p \mid A$.

Let $w_1, \dots, w_e$ be the conjugates of w . Since $\mathbb{Q}(w) = \mathbb{Q}(\beta)$, the multiset $\{w_i^p\}$ is exactly the multiset of conjugates of β . Since G/u is monic over $\mathbb{Z}_{(p)}$, the w_i are integral over $\mathbb{Z}_{(p)}$. In the integral closure $\mathcal{O}$ of $\mathbb{Z}_{(p)}$,

$$\begin{aligned} G(x)^p &= u^p \prod_i (x - w_i)^p \\ &\equiv u^p \prod_i (x^p - w_i^p) = h(x^p) \pmod{p\mathcal{O}[x]}. \end{aligned}$$

Both sides have rational coefficients, and $p\mathcal{O} \cap \mathbb{Q} = p\mathbb{Z}_{(p)}$, so this is a coefficientwise congruence in $\mathbb{Z}_{(p)}[x]$. Coefficientwise freshman's dream also gives $G(x)^p \equiv G(x^p)$ and $h(x)^p \equiv h(x^p) \pmod{p}$. Hence $\overline{G(x^p)} = \overline{h(x^p)}$; injectivity of $f(x) \mapsto f(x^p)$ on $\mathbb{F}_p[x]$ yields

$$\overline{G} = \overline{h}.$$

Define

$$\Phi_p(f)(x) = \frac{f(x)^p - f(x^p)}{p}.$$

Because u is a p -adic unit, the quotient

$$R = \mathbb{Z}_{(p)}[x]/(G) = \mathbb{Z}_{(p)}[x]/(G/u)$$

is a free, hence torsion-free, $\mathbb{Z}_{(p)}$ -module of rank e . Since $\overline{G} = \overline{h}$, write $G = h + pE$ with $E \in \mathbb{Z}_{(p)}[x]$. In R , one has

$$h = -pE, \quad h(x^p) = 0,$$

the second equality following from $G \mid h(x^p)$. Therefore

$$\Phi_p(h) = (-1)^p p^{p-1} E^p \in pR.$$

Since $R/pR \simeq \mathbb{F}_p[x]/(\overline{h})$, this proves

$$\overline{h} \mid \overline{\Phi_p(h)}.$$

The product rule

$$\Phi_p(fg) = f^p \Phi_p(g) + g(x^p) \Phi_p(f)$$

transfers this vanishing from h to F . Applying the same rule to

$$H(x) = (x-1)^2 F(x) = Ax^N - Nx^A + B$$

transfers it from F to H : at a root α of $\overline{h}$, the second product-rule term contains $F(\alpha^p) = F(\alpha)^p = 0$ in characteristic p . Thus

$$\overline{\Phi_p(H)}(\alpha) = 0 \quad \text{for every root } \alpha \text{ of } \overline{h}.$$

Set

$$U = A\alpha^B, \quad V = B\alpha^{-A}.$$

The equation $H(\alpha) = 0$ gives $U + V = N$. Put

$$q_p(t) = \frac{t^p - t}{p}, \quad C_p(X, Y) = \frac{(X+Y)^p - X^p - Y^p}{p}.$$

To make the coefficient calculation explicit, write

$$H = Ax^N - Nx^A + B.$$

In $\Phi_p(H)$, the mixed terms from the three summands contribute $x^{pA}C_p(U, V)$, while the coefficient corrections contribute

$$x^{pA} \left(\frac{q_p(A)}{A} U^p + \frac{q_p(B)}{B} V^p - q_p(N) \right).$$

Using $V = N - U$ gives

$$\alpha^{-pA} \overline{\Phi_p(H)}(\alpha) = T_{p,A,B}(U), \tag{14.6}$$

where

$$T_{p,A,B}(U) = C_p(U, N - U) + \frac{q_p(A)}{A}U^p + \frac{q_p(B)}{B}(N - U)^p - q_p(N) \in \mathbb{F}_p[U]. \quad (14.7)$$

The hypothesis $p \nmid ABN$ is essential here. The polynomial has degree at most p , and the identity

$$q_p(N) = q_p(A) + q_p(B) + C_p(A, B)$$

gives

$$T_{p,A,B}(A) = T'_{p,A,B}(A) = 0, \quad T''_{p,A,B}(A) = A^{-1} + B^{-1} = \frac{N}{AB} \neq 0.$$

Thus $T_{p,A,B}$ is nonzero and $U = A$ is an exactly double root.

It remains to count the other roots. The reduction of F is square-free. Indeed, a common root of F and F' would be a common root of H and

$$H'(x) = ANx^{A-1}(x^B - 1).$$

It would satisfy $x^A = x^B = 1$, hence $x = 1$, whereas

$$F(1) = \frac{ABN}{2} \not\equiv 0 \pmod{p}.$$

Therefore $\bar{h}$ has e distinct roots. The map

$$\alpha \mapsto U = A\alpha^B$$

is injective on roots of F : equality of U also gives equality of $V = N - U$, so the quotient of two roots has both its A -th and B -th powers equal to 1. Moreover $U = A$ would force $\alpha^A = \alpha^B = 1$, hence $\alpha = 1$, which is not a root. Consequently the e roots of $\bar{h}$ give e distinct roots of $T_{p,A,B}$ outside the exactly double root A . The remaining root-multiplicity budget is at most $p - 2$, proving (14.3). $\square$

Theorem 14.6 (Self-power rigidity). *Let A, B be coprime positive integers with $A \neq B$, and suppose $F = Q_{A,B}$ is irreducible. If β is a root and $K = \mathbb{Q}(\beta)$, then*

$$\beta \notin K^p \quad \text{for every prime } p. \quad (14.8)$$

Proof. Suppose $\beta = w^p$ with $w \in K$. Put $N = A + B$ and $D = N - 2$. Since the monic minimal polynomial of β is F/A ,

$$N_{K/\mathbb{Q}}(w)^p = N_{K/\mathbb{Q}}(\beta) = (-1)^D \frac{B}{A}. \quad (14.9)$$

For odd p , coprimality implies

$$A = u^p, \quad B = v^p$$

for coprime positive integers u, v . For $p = 2$, equation (14.9) forces D even and A, B to be coprime squares; hence both are odd and $v_2(N) = 1$.

First suppose $p \mid ABN$. For odd p , Lemma 14.4 provides in each of the three possible roles a cluster at 1 containing exactly $e = p - 2$ roots γ with

$$v_p(\gamma - 1) = \frac{1}{e}.$$

For $p = 2$, the preceding parity conclusion gives $v_2(N) = 1$. Since A, B are distinct coprime odd squares, $N \geq 10$, so $N/2 > 1$; Borisov's Lemma 2.8 provides a nontrivial prime-to-2 center ζ with exactly $e = 2$ roots satisfying $v_2(\gamma - \zeta) = 1/2$.

Because F is irreducible, a $\mathbb{Q}$ -embedding of K into $\overline{\mathbb{Q}}_p$ may send β to one of these roots. Let L be the unramified field generated by the cluster center. Every L -embedding fixes the cluster center and preserves the unique extension of the p -adic valuation, so $v_p(\sigma(\beta) - \zeta) = v_p(\beta - \zeta)$. Thus every L -conjugate remains on the same edge, which contains only e roots; hence $[L(\beta) : L] \leq e$. Lemma 14.3 contradicts $\beta = w^p$.

It remains that $p \nmid ABN$. Then p is odd, and Theorem 14.5, applied to the whole irreducible polynomial of degree D , gives $D \leq p - 2$. But $A = u^p$, $B = v^p$, and $u \neq v$, so

$$D = A + B - 2 \geq 1 + 2^p - 2 = 2^p - 1 > p - 2,$$

a contradiction. This proves (14.8). $\square$

14.2 Kummer reduction and its complete elimination

Lemma 14.7 (Degree under extraction of a root). *Let $f \in \mathbb{Q}[x]$ be irreducible of degree D , let $r \geq 1$, and let w be algebraic with $f(w^r) = 0$. Put $\beta = w^r$. Then*

$$[\mathbb{Q}(w) : \mathbb{Q}] = [\mathbb{Q}(w) : \mathbb{Q}(\beta)]D.$$

In particular, equality of the two degrees is equivalent to $\mathbb{Q}(w) = \mathbb{Q}(\beta)$.

Proof. This is the tower law for $\mathbb{Q}(\beta) \subseteq \mathbb{Q}(w)$, together with the irreducibility of f . $\square$

Theorem 14.8 (Exact Kummer reduction). *Let A, B be coprime and unequal, put $N = A + B$, and suppose $Q_{A,B}$ is irreducible. Assume that $\{rA, rB\}$ shares a factor with $\{sC, sD\}$, where $\gcd(C, D) = 1$. Independently interchange A, B and C, D , as necessary, so that the selected common root lies in $Q_{rA, rB}$ and $Q_{sC, sD}$. Put*

$$g = \gcd(r, s), \quad r_0 = r/g, \quad s_0 = s/g.$$

Then either the two unordered packages agree, or $r_0 > 1$ and a root β of $Q_{A,B}$ is an r_0 -th power inside $\mathbb{Q}(\beta)$.

Proof. Let α be the common root and set $w = \alpha^g$. Then

$$Q_{A,B}(w^{r_0}) = 0, \quad Q_{C,D}(w^{s_0}) = 0.$$

Let k_w be the minimal polynomial of w , and put $M = C + D$. Because $|w| = |\alpha|^g \neq 1$, the polynomial k_w is noncyclotomic. It divides both normalized package polynomials

$$Q_{A,B}(x^{r_0}), \quad Q_{C,D}(x^{s_0}).$$

The associated exponent-pair gcds are r_0, s_0 , which are coprime. Theorem 14.2 gives

$$\deg k_w \leq \min(N, M) - 2 \leq N - 2.$$

On the other hand, $\beta = w^{r_0}$ is a root of the irreducible polynomial $Q_{A,B}$ of degree $N - 2$, so Lemma 14.7 gives the reverse inequality. Hence $\mathbb{Q}(w) = \mathbb{Q}(\beta)$.

If $r_0 = 1$, then $Q_{A,B}$ divides Q_{s_0C, s_0D} , and Theorem 13.2 forces equality of the oriented pairs and hence of the packages. If $r_0 > 1$, the field equality gives $w \in \mathbb{Q}(\beta)$ and $w^{r_0} = \beta$, as asserted. $\square$

Theorem 14.9 (Global isolation of every lift of an irreducible shape). *Let A, B be coprime and unequal, and suppose $Q_{A,B}$ is irreducible. Then for every $r \geq 1$, the package $\{rA, rB\}$ is globally isolated.*

Proof. An unequal-package collision would give the second alternative in Theorem 14.8. Choose a prime $p \mid r_0$; then $\beta = (w^{r_0/p})^p$ inside $\mathbb{Q}(\beta)$, contradicting Theorem 14.6. $\square$

Corollary 14.10 (Individual composition irreducibility also isolates). *Let A, B be coprime and unequal, and $r \geq 1$. If $Q_{A,B}(x^r)$ is irreducible over $\mathbb{Q}$, then $\{rA, rB\}$ is globally isolated.*

Proof. If a common factor existed, a source root α would have degree $r(A + B - 2)$. With the notation in the preceding proof, $[\mathbb{Q}(\alpha) : \mathbb{Q}(w)] \leq g$, so

$$[\mathbb{Q}(w) : \mathbb{Q}] \geq r_0(A + B - 2).$$

The Schinzel bound is at most $A + B - 2$, forcing $r_0 = 1$; proper-divisibility rigidity then forces equality of the packages. $\square$

Theorem 14.11 (Density-one global isolation). *Let $I(X)$ denote the number of unordered pairs $1 \leq a < b$, $a + b \leq X$, whose reciprocal packages are not globally isolated. Then*

$$I(X) \ll X^{20/11} \log(2X). \quad (14.10)$$

Proof. Write uniquely

$$a = rA, \quad b = rB, \quad \gcd(A, B) = 1, \quad A < B.$$

By Theorem 14.9, a non-isolated package must have reducible primitive base $Q_{A,B}$.

To connect precisely with Borisov's notation, his triple is

$$a_{\text{Bor}} = A + B, \quad b_{\text{Bor}} = A, \quad c_{\text{Bor}} = B.$$

For a fixed small $\varepsilon > 0$, the contrapositive of his Theorem 3.8 places every sufficiently large reducible primitive shape in the complementary numerical range counted by Theorem 3.9 [1]. The finitely many small totals are absorbed into the constant. Thus the number $R(Y)$ of reducible primitive shapes of total at most Y satisfies

$$R(Y) \ll Y^{20/11} \log(2Y). \quad (14.11)$$

Therefore

$$I(X) \leq \sum_{r \leq X/3} R(X/r) \ll X^{20/11} \log(2X) \sum_{r \geq 1} r^{-20/11} \ll X^{20/11} \log(2X).$$

Since the total number of unordered unequal pairs of total at most X is asymptotic to $X^2/4$, a density-one set of package vertices is globally isolated. $\square$

Remark 14.12. The estimate is global, not a bounded-box assertion: every certified package is coprime to every distinct package in the entire infinite family. The remaining set is merely the set not eliminated by the present proof; the theorem does not assert that its members actually share factors.

15 Factor-level power sieves, residual splitting, and exact signatures

The local cluster argument constrains proper factors without assuming irreducibility of the whole primitive polynomial. We first retain the radical sieve, then resolve the equal-slope 2-adic block by its residual polynomial.

Theorem 15.1 (Primitive total-radical sieve). *Let A, B be coprime positive integers, put $N = A+B$, and let $h \in \mathbb{Q}[x]$ be an irreducible factor of $Q_{A,B}$ of degree e . For every prime $p \mid N$,*

$$e \equiv 0 \quad \text{or} \quad -2 \pmod{p}. \quad (15.1)$$

Consequently,

$$\text{rad}(N) \mid e(e+2). \quad (15.2)$$

If the same reciprocal irreducible orbit occurs in primitive packages of totals N and M , then

$$\text{lcm}(\text{rad } N, \text{rad } M) \mid e(e+2). \quad (15.3)$$

Proof. Fix $p^k \parallel N$. Since $p \nmid AB$, normalize a rational factor to be monic over $\mathbb{Z}_p$ with unit constant term, and work over the unramified extension containing the prime-to- p cluster centers.

Assume first that p is odd. At every nontrivial center, Borisov's Lemma 2.8 gives nonhorizontal Newton edges of lengths

$$p, \quad p^2 - p, \quad \dots, \quad p^k - p^{k-1},$$

with reduced slope denominators equal to those lengths. At the center 1, Lemma 14.4 gives a side of length $p - 2$ and reduced slope denominator $p - 2$; Borisov's Lemma 2.9 gives the subsequent p -divisible lengths. A monic local factor has a Newton polygon with integral vertices. Its horizontal contribution on an edge of reduced slope denominator ℓ is divisible by ℓ ; here ℓ equals the full edge length, so the factor takes that edge wholly or not at all. Contributions at nontrivial centers are divisible by p , while the center 1 contributes either 0 or $p - 2$ modulo p . This proves (15.1) for odd p .

For $p = 2$, one must not separate Borisov's first two formal layers when $k \geq 2$: both have slope $1/2$ and form a single canonical side of length four. Its reduced slope denominator is nevertheless 2, so any local factor takes an even number of roots from that side. Every later nontrivial side has even reduced slope denominator, and the same is true for every nonhorizontal side at the center 1. Hence every factor degree is even. This is precisely (15.1) modulo 2.

Thus every $p \mid N$ divides $e(e+2)$, proving (15.2). For a common reciprocal orbit, apply the result to whichever of $h, h^\vee$ divides the second orientation; the reciprocal has the same degree. This gives (15.3). $\square$

Corollary 15.2 (A numerical coprimality criterion). *Let two primitive packages have totals N, M , put*

$$\nu = \min(N, M), \quad \rho = \text{lcm}(\text{rad } N, \text{rad } M).$$

If

$$\rho > \nu(\nu - 2), \quad (15.4)$$

then the packages are coprime. In particular, distinct primitive packages with coprime squarefree totals are coprime.

Proof. A common factor has degree $1 \leq e \leq \nu - 2$, so $e(e+2) \leq \nu(\nu - 2)$, contradicting (15.3)–(15.4). For coprime squarefree totals, $\rho = NM > \nu(\nu - 2)$. $\square$

15.1 The merged 2-adic side is residual-separable

We use the first-order theorem of the residual polynomial in the following standard form; see [9, Theorem 1.21].

Lemma 15.3 (Newton–Ore residual splitting). *Let $L/\mathbb{Q}_2$ be unramified and let a polynomial over L have a Newton side of slope $-1/2$, horizontal length four, and separable quadratic residual polynomial. After a finite unramified extension L'/L , the four roots belonging to that side split into two packets of two. Each root γ in either packet satisfies*

$$[L'(\gamma) : L'] \leq 2.$$

Proof. The denominator of the reduced slope is 2, and the side has residual degree two. After an unramified extension splitting the separable residual polynomial, its two distinct linear residual factors give, by the theorem of the residual polynomial, two factors of degree 2. Each associated root therefore has local degree at most two over the enlarged unramified base. $\square$

Proposition 15.4 (Residual polynomials of the merged block). *Let A, B be coprime, $N = A + B$, and suppose a nontrivial prime-to-2 cluster center exists.*

(a) *If $N = 2^\kappa n$, $\kappa \geq 2$, n odd, and $\zeta \neq 1$ is an n -th root of unity, then the length-four slope-1/2 side at ζ has residual polynomial*

$$R_N(Z) = (1 - \bar{\zeta}^A) + \bar{\zeta}^{-2}Z + \bar{\zeta}^{-4}Z^2.$$

(b) *If $B = 2^\kappa b$, $\kappa \geq 2$, b odd, and $\zeta \neq 1$ is a b -th root of unity, then the corresponding unit side has residual polynomial*

$$R_B(Z) = (1 - \bar{\zeta}^A) + \bar{\zeta}^{A-2}Z + \bar{\zeta}^{A-4}Z^2.$$

Both polynomials are separable. The case $2^\kappa \parallel A$ follows by reciprocity.

Proof. Work in the unramified field generated by ζ , and write

$$H_{A,N}(\zeta + Y) = \sum_{j \geq 0} c_j Y^j.$$

At a nontrivial center the denominator $(x-1)^2$ is a unit, so the relevant Newton data are unchanged on passing to $Q_{A,B}$.

In case (a), $\zeta^N = 1$, A is odd, and

$$\begin{aligned} c_0 &= N(1 - \zeta^A), \\ c_2 &= \frac{AN}{2} \zeta^{-2} ((N-1) - (A-1)\zeta^A), \\ c_4 &= A \binom{N}{4} \zeta^{-4} - N \binom{A}{4} \zeta^{A-4}. \end{aligned}$$

The three valuations are respectively $\kappa, \kappa-1, \kappa-2$. After division by the powers of two on the side and reduction, the coefficients are

$$1 - \bar{\zeta}^A, \quad \bar{\zeta}^{-2}, \quad \bar{\zeta}^{-4}.$$

Here the second term in c_4 has larger valuation, while $2^{2-\kappa}\binom{N}{4}$ is odd.

In case (b), $\zeta^B = 1$, and direct cancellation gives

$$c_0 = B(1 - \zeta^A), \quad c_2 = \frac{ANB}{2}\zeta^{A-2},$$

$$c_4 = \frac{ANB}{24}(N^2 + NA + A^2 - 6(N + A) + 11)\zeta^{A-4}.$$

Since A, N are odd and $N \equiv A \pmod{4}$, the integer in parentheses is congruent to 2 (mod 4). The normalized residual coefficients are therefore

$$1 - \bar{\zeta}^A, \quad \bar{\zeta}^{A-2}, \quad \bar{\zeta}^{A-4}.$$

Borisov's polygon identifies these points as the merged canonical side in both cases. In characteristic two,

$$R'_N(Z) = \bar{\zeta}^{-2} \neq 0, \quad R'_B(Z) = \bar{\zeta}^{A-2} \neq 0.$$

Thus both residual polynomials are separable. $\square$

Corollary 15.5 (Internal powers at every total prime). *Let h be an irreducible factor of a primitive $Q_{A,B}$, and let β be a root. If $p \mid A + B$, then*

$$\beta \notin \mathbb{Q}(\beta)^p.$$

This includes $p = 2$ for every value of $v_2(A + B)$.

Proof. For odd p , a root lies on a Newton side of some length ℓ and slope denominator ℓ . Borisov's exact count gives at most ℓ roots at that center and distance, while the valuation denominator forces the local ramification degree to be at least ℓ . Lemma 14.3 applies.

For $p = 2$, sides of length two with no equal-slope merger are handled in the same way. If the primitive total has odd part greater than one and $v_2(A + B) \geq 2$, Proposition 15.4(a) and Lemma 15.3 split the four-root side, after an unramified extension, into two degree-two packets. Lemma 14.3 then applies to each packet. If the primitive total is a pure power of two, there are no nontrivial centers; at the center 1, the nonzero sides have successive lengths 2, 4, 8, ... and distinct slopes, so the original edge argument applies directly. These cases exhaust all roots. $\square$

15.2 Endpoint and good-prime power sieves

Theorem 15.6 (Endpoint-prime internal-power sieve). *Let h be an irreducible factor of primitive $Q_{A,B}$, of degree e , and let β be a root.*

(a) *Suppose $p^\kappa \parallel B$, for an arbitrary prime p . If $\beta \in \mathbb{Q}(\beta)^p$, then*

$$e \leq A, \quad pA \mid e\kappa. \quad (15.5)$$

(b) *Suppose $p^\kappa \parallel A$. Then an internal p -th power forces*

$$e \leq B, \quad pB \mid e\kappa. \quad (15.6)$$

Equivalently, in (a),

$$\frac{pA}{\gcd(pA, \kappa)} \mid e;$$

in particular $\kappa < p$ is impossible. The reciprocal statement holds in (b).

Proof. We prove (a); reciprocity gives (b). Borisov's endpoint description [1, Lemma 2.10] gives exactly A nonunit roots, each of valuation κ/A . Every other root lies in a prime-to- p root-of-unity cluster; the reciprocal statement uses [1, Lemma 2.11].

For odd p , every unit-cluster side has exact root count equal to its reduced slope denominator, so Lemma 14.3 excludes an internal p -th power there. For $p = 2$, the unmerged sides are treated identically. If $\kappa \geq 2$ and the odd part of B is greater than one, Proposition 15.4(b) and Lemma 15.3 split the merged side into degree-two packets. If the odd part is one, there are no nontrivial centers and the center-1 sides have distinct slopes. Thus unit-cluster roots are excluded for every p and every κ .

If $\beta = w^p$, every rational conjugate of β is likewise a p -th power in its conjugate field. Hence all e roots of h must belong to the nonunit block, proving $e \leq A$. The leading coefficient of h is a p -adic unit, and therefore

$$v_p\left(N_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta)\right) = \frac{e\kappa}{A}.$$

The norm is a rational p -th power, so this integer is divisible by p . Thus $pA \mid e\kappa$. If $\kappa < p$, then $e\kappa \leq A\kappa < pA$, a contradiction. $\square$

Corollary 15.7 (Good-prime internal-power sieve). *Under the hypotheses of Theorem 14.5,*

$$\text{rad}(A + B) \leq e(e + 2) \leq p(p - 2).$$

In particular, $\text{rad}(A + B) > p(p - 2)$ rules out an internal p -th power for every factor.

Proof. Combine Theorems 15.1 and 14.5. $\square$

We next refine the Kummer trigger prime by prime. We use the prime-degree case of Capelli's binomial criterion [10, Chapter VI, Section 9]: over a characteristic-zero field K , the polynomial $x^p - a$, with p prime, is irreducible unless $a \in K^p$.

Proposition 15.8 (Prime-by-prime Kummer trigger). *Consider selected orientations of packages with primitive shapes (A, B) and (C, D) , scaling factors r, s , and a common root α . Put*

$$g = \gcd(r, s), \quad r_0 = r/g, \quad s_0 = s/g, \quad w = \alpha^g, \quad L = \min(A + B, C + D) - 2.$$

Let $\beta = w^{r_0}$ be a root of an irreducible factor $h \mid Q_{A,B}$ of degree e . For every prime $p \mid r_0$, at least one of the following holds:

- (i) $pe \leq L$;
- (ii) $\beta \in \mathbb{Q}(\beta)^p$.

If $pe > L$, the second alternative holds and hence all applicable conclusions of Corollary 15.5, Theorem 15.6, and Theorem 14.5 apply.

Proof. The minimal polynomial k_w is noncyclotomic because $|w| = |\alpha|^g \neq 1$. It divides $Q_{A,B}(x^{r_0})$ and $Q_{C,D}(x^{s_0})$, whose exponent-pair gcds are coprime. Theorem 14.2 gives

$$[\mathbb{Q}(w) : \mathbb{Q}] \leq L.$$

Fix $p \mid r_0$, put $u = w^{r_0/p}$, and set $K = \mathbb{Q}(\beta)$. If $\beta \notin K^p$, Capelli's criterion makes $x^p - \beta$ irreducible over K , so

$$[K(u) : K] = p.$$

Since $K(u) \subseteq \mathbb{Q}(w)$, it follows that $pe \leq [\mathbb{Q}(w) : \mathbb{Q}] \leq L$. The contrapositive gives the second alternative when $pe > L$. $\square$

Remark 15.9. The former condition $2e > L$ is the special case in which the internal-power alternative is forced simultaneously for every prime divisor of r_0 . The prime-by-prime form is stronger when the relative scaling exponent has a large prime divisor.

15.3 Role assignments and exact factor partitions

Proposition 15.10 (Odd-prime degree signatures). *Suppose a reciprocal irreducible orbit of degree e occurs in primitive packages of totals N and M . Put*

$$\rho_{\text{odd}} = \text{lcm}(\text{rad } N, \text{rad } M)_{\text{odd}}.$$

Then there is a unique factorization

$$\rho_{\text{odd}} = UV, \quad (U, V) = 1, \quad U \mid e, \quad V \mid e + 2, \quad (15.7)$$

obtained by assigning each odd prime divisor of ρ_{odd} to exactly one of the two roles $e \equiv 0$ or $e \equiv -2$. If $2 \mid NM$, then e is even.

Proof. For every odd $q \mid \rho_{\text{odd}}$, Theorem 15.1 says that $q \mid e(e + 2)$. Since $\gcd(e, e + 2) \mid 2$, the prime q divides exactly one of the two integers. Multiplying the primes in their two roles gives (15.7). The assertion at 2 is the parity part of the same theorem. $\square$

Corollary 15.11 (Exact factor-partition signature). *Let $N = qk$, where q is an odd prime, and let the irreducible factor degrees of a primitive $Q_{A,B}$ be $e_1, \dots, e_t$. Write uniquely*

$$e_i = qa_i - 2\varepsilon_i, \quad a_i \geq 1, \quad \varepsilon_i \in \{0, 1\},$$

and put $t_q = \sum_i \varepsilon_i$. Then for an integer $j \geq 0$,

$$t_q = 1 + qj, \quad \sum_i a_i = k + 2j, \quad (q - 2)j \leq k - 1. \quad (15.8)$$

In particular, if $q > k + 1$, then $j = 0$: exactly one factor has the -2 -role at q , and the total number of irreducible factors is at most $k = N/q$. If $N = 2q$ with $q \geq 5$, the q -signature alone would force any reducible factorization to have two factors of degrees $q - 2$ and q .

Proof. Summing the factor degrees gives

$$q \sum_i a_i - 2t_q = qk - 2.$$

Reduction modulo q yields $t_q \equiv 1 \pmod{q}$, so $t_q = 1 + qj$, and substitution gives $\sum_i a_i = k + 2j$. Since each index counted by t_q contributes at least one to $\sum_i a_i$,

$$1 + qj = t_q \leq \sum_i a_i = k + 2j,$$

which is the final inequality in (15.8). If $q > k + 1$, then $q - 2 > k - 1$, forcing $j = 0$. For $k = 2$, reducibility forces two positive a_i 's summing to two, and exactly one ε_i is one; the degrees are therefore $q - 2$ and q . $\square$

Corollary 15.12 (Prime and twice-prime totals). *Let A, B be coprime and unequal. If $A + B = q$, or if $A + B = 2q$, where q is an odd prime, then $Q_{A,B}$ is irreducible over $\mathbb{Q}$. Consequently every positive integral lift of such a primitive shape is globally isolated.*

Proof. If the total is q , Theorem 15.1 says that a factor degree e , with $1 \leq e \leq q - 2$, is congruent to 0 or $-2 \pmod{q}$. The only positive possibility in this range is $e = q - 2$, the full degree.

Now let the total be $2q$. The prime 2 in the radical sieve makes every factor degree even. The q -congruence shows that a proper factor degree in the range $1 \leq e < 2q - 2$ can only be $q - 2$ or q , both odd. Thus no proper factor exists. Global isolation of all lifts follows from Theorem 14.9. $\square$

The following role rigidity is useful when a common factor is much smaller than a prime in the shared additive-triple support.

Proposition 15.13 (Large-prime endpoint-role rigidity). *Let an irreducible polynomial h of degree e divide selected primitive orientations $Q_{A,B}$ and $Q_{C,D}$. Let $p \geq 5$ be prime with $p > e + 2$, and suppose $p^\kappa \parallel B$. Then $p \nmid C + D$, and necessarily $p^\lambda \parallel D$ for some $\lambda \geq 1$. Moreover*

$$\frac{\kappa}{A} = \frac{\lambda}{C}, \quad \frac{A}{\gcd(A, \kappa)} \mid e, \quad \frac{C}{\gcd(C, \lambda)} \mid e. \quad (15.9)$$

In particular, $\kappa = \lambda = 1$ is impossible for two distinct packages.

Proof. Support compatibility, Corollary 11.5, forces $p \mid CD(C + D)$. It cannot divide $C + D$, because the radical sieve would give $e \equiv 0$ or $-2 \pmod{p}$, impossible when $0 < e < p - 2$.

At an endpoint prime, every unit-cluster side has reduced slope denominator at least $p - 2 > e$. A local factor of the degree- e polynomial h cannot contain such a root. Hence every root of h belongs to the nonunit block. In $Q_{A,B}$ these valuations are positive and equal to κ/A ; therefore the second orientation must also put p in its constant endpoint, namely D , rather than in its leading endpoint C , whose nonunit valuations have the opposite sign. Equality of the valuations gives the first relation in (15.9). Integrality of the valuation of the norm gives the two denominator divisibilities.

If $\kappa = \lambda = 1$, then $A \mid e$ and $C \mid e$. The two nonunit blocks also give $e \leq A, C$, so $A = C = e$, contradicting the disjointness of the two additive triples. $\square$

The merged 2-adic slope is therefore no longer an unresolved local case. The remaining bad-endpoint room is precise: an internal p -th power must lie entirely in a nonunit endpoint block and requires an endpoint valuation exponent at least p , together with the divisibilities in (15.5)–(15.6).

16 Reciprocal resultants as a supplementary sieve

For primitive a, b , put $[k] = 1 + x + \cdots + x^{k-1}$. Reciprocal polynomials are normalized as in Section 4. One has

$$Q_{a,b}(x) + Q_{b,a}(x) = (a + b)[a](x)[b](x). \quad (16.1)$$

Proposition 16.1 (Exact reciprocal resultant). *If $\gcd(a, b) = 1$, then*

$$|\text{Res}(Q_{a,b}, Q_{b,a})| = a^{a-2}b^{b-2}(a + b)^{a+b-2}, \quad (16.2)$$

where a factor 1^{-1} is interpreted as 1.

Proof. At a root α of $Q_{a,b}$, equation (16.1) gives

$$Q_{b,a}(\alpha) = (a + b)[a](\alpha)[b](\alpha).$$

At a nontrivial a -th root of unity ζ ,

$$Q_{a,b}(\zeta) = a \frac{\zeta^b - 1}{(\zeta - 1)^2}.$$

Because multiplication by b permutes the nontrivial a -th roots,

$$|\text{Res}(Q_{a,b}, [a])| = a^{a-2}.$$

Similarly, $|\text{Res}(Q_{a,b}, [b])| = b^{b-2}$. More explicitly,

$$\begin{aligned} \text{Res}(Q_{a,b}, Q_{b,a}) &= \text{Res}(Q_{a,b}, (a+b)[a][b] - Q_{a,b}) \\ &= (a+b)^{a+b-2} \text{Res}(Q_{a,b}, [a]) \text{Res}(Q_{a,b}, [b]). \end{aligned}$$

The preceding two evaluations give the formula. $\square$

If $a = gA$, $b = gB$, $\gcd(A, B) = 1$, then composition and scaling give

$$|\text{Res}(Q_{a,b}, Q_{b,a})| = g^{2g(A+B-2)} \left(A^{A-2} B^{B-2} (A+B)^{A+B-2} \right)^g. \quad (16.3)$$

These formulas are useful as a sieve for a proposed common irreducible factor, but cannot by themselves prove coprimality: a nonreciprocal irreducible polynomial may have reciprocal resultant 1.

17 Sparse Wronskians, explicit absolute boundedness, and arithmetic sieves

Let

$$W_{m,n;p,q}(x) = H'_{m,n}(x)H_{p,q}(x) - H_{m,n}(x)H'_{p,q}(x).$$

Proposition 17.1 (Sparse Wronskian). *If a noncyclotomic irreducible polynomial h divides both $H_{m,n}$ and $H_{p,q}$, then*

$$h(x)^2 \mid \frac{W_{m,n;p,q}(x)}{(x-1)^4}. \quad (17.1)$$

Moreover the numerator has at most eight monomials before equal exponents are merged:

$$\begin{aligned} W_{m,n;p,q}(x) &= mp(n-q)x^{n+q-1} + mq(p-n)x^{n+p-1} \\ &\quad + np(q-m)x^{m+q-1} + nq(m-p)x^{m+p-1} \\ &\quad + mn(q-p)(x^{n-1} - x^{m-1}) \\ &\quad - pq(n-m)(x^{q-1} - x^{p-1}). \end{aligned} \quad (17.2)$$

If the ordered edges are distinct, then $W_{m,n;p,q} \not\equiv 0$.

Proof. Write $H_{m,n} = (x-1)^2 K_1$ and $H_{p,q} = (x-1)^2 K_2$. Then

$$W_{m,n;p,q} = (x-1)^4 (K'_1 K_2 - K_1 K'_2).$$

If $K_1 = hA$ and $K_2 = hB$, the terms containing h' cancel:

$$K'_1 K_2 - K_1 K'_2 = h^2 (A'B - AB').$$

This proves (17.1), and direct expansion gives (17.2). If the Wronskian vanished identically, then $(H_{m,n}/H_{p,q})' = 0$, so the two trinomials would be scalar multiples. Comparison of their three nonzero supports $\{0, m, n\}$ and $\{0, p, q\}$ would force $(m, n) = (p, q)$. $\square$

The Wronskian supplies a square-factor obstruction for each fixed exponent tuple. A uniform structural reduction comes from applying the low-height sparse-gcd theorem of Amoroso–Sombra–Zannier directly to the two collision trinomials. We first make the two geometric inputs completely formal.

Lemma 17.2 (Four-unit divisor rigidity). *Let C be a smooth projective curve over an algebraically closed field K of characteristic zero, and let $u, v \in K(C)$ be nonconstant. If*

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{\operatorname{div}(u), \operatorname{div}(u-1), \operatorname{div}(v), \operatorname{div}(v-1)\} \leq 2,$$

then

$$v \in \left\{ u, 1-u, u^{-1}, (1-u)^{-1}, \frac{u}{u-1}, \frac{u-1}{u} \right\}.$$

Proof. Put

$$\Delta_0 = \operatorname{div}(u), \quad \Delta_1 = \operatorname{div}(u-1), \quad E_0 = \operatorname{div}(v), \quad E_1 = \operatorname{div}(v-1).$$

A nonconstant function on a smooth projective curve is surjective onto $\mathbb{P}^1$. Hence the fibres of u over $0, 1, \infty$ are all nonempty. The divisors Δ_0, Δ_1 are linearly independent: at a point over $u = 0$, only Δ_0 has positive valuation, and at a point over $u = 1$, only Δ_1 does. Thus

$$E_0 = a\Delta_0 + b\Delta_1, \quad E_1 = c\Delta_0 + d\Delta_1$$

for rational a, b, c, d . In particular, the zero, one, and pole fibres of v are supported over the three special fibres of u . Moreover the role is uniform on each such fibre. Indeed, at a point P over $u = 0$,

$$(\operatorname{ord}_P(v), \operatorname{ord}_P(v-1)) = \operatorname{ord}_P(u)(a, c),$$

while over $u = 1$ it is $\operatorname{ord}_P(u-1)(b, d)$, and over $u = \infty$ it is $-\operatorname{ord}_P(u^{-1})(a+b, c+d)$. The coefficient pair is independent of the chosen point in that fibre.

Because $v - (v-1) = 1$, the normalized valuation pair on each special fibre is therefore uniformly one of

$$(r, 0), \quad (0, r), \quad (-r, -r) \quad (r > 0), \quad \text{or} \quad (0, 0).$$

Since the nonconstant function v assumes all three values $0, 1, \infty$, the three nonneutral roles occur, and they occur bijectively on the three special fibres of u . Replacing v by one of the six Möbius transformations permuting $\{0, 1, \infty\}$ preserves the same two-dimensional divisor span. We may therefore arrange that the zero, one, and pole fibres of v lie respectively over those of u . Then, for positive rationals r_0, r_1 ,

$$\operatorname{div}(v) = r_0 \operatorname{div}(u), \quad \operatorname{div}(v-1) = r_1 \operatorname{div}(u-1).$$

At the common pole fibre, v and $v-1$ have the same negative valuation, so $r_0 = r_1 =: r$.

Write $r = A/B$ with A, B positive coprime integers. The divisor equalities give constants $\kappa, \lambda \neq 0$ such that

$$v^B = \kappa u^A, \quad (v-1)^B = \lambda (u-1)^A.$$

Logarithmic differentiation gives

$$B \frac{dv}{v} = A \frac{du}{u}, \quad B \frac{dv}{v-1} = A \frac{du}{u-1}.$$

In characteristic zero, $du \neq 0$, and hence also $dv \neq 0$. Comparing the two identities gives

$$\frac{v}{u} = \frac{v-1}{u-1},$$

so $v = u$. Undoing the initial Möbius permutation gives the six possibilities. $\square$

Proposition 17.3 (Rank-two torus-intersection rigidity). *Let $0 < m < n$, $0 < p < q$, assume*

$$\gcd(\gcd(m, n), \gcd(p, q)) = 1,$$

and suppose the two unordered packages are distinct. Put

$$X = V((n - m) - nx_1 + mx_2, (q - p) - qx_3 + px_4) \subset \mathbb{G}_m^4.$$

There is no geometric irreducible curve contained in the intersection of X with a two-dimensional subtorus and containing a point

$$(\xi^m, \xi^n, \xi^p, \xi^q)$$

for which $\xi \notin \mu_\infty$ is a common root of $H_{m,n}$ and $H_{p,q}$.

Proof. Base-change to an algebraic closure and work on the smooth projective model of the normalization of such a curve C . Introduce

$$\begin{aligned} u &= \frac{n}{n-m}x_1, & u-1 &= \frac{m}{n-m}x_2, \\ v &= \frac{q}{q-p}x_3, & v-1 &= \frac{p}{q-p}x_4. \end{aligned}$$

Suppose first that one projection of C to a punctured factor line is constant. The other projection is then nonconstant and hence dominant. A monomial $x_3^a x_4^b$ constant on the second line must have $a = b = 0$: parameterize the line by

$$x_3 = t, \quad x_4 = \frac{qt - (q-p)}{p},$$

and compare the valuations at $t = 0$, $t = (q-p)/q$, and infinity. Thus, if the first projection is constant, the two independent characters of the containing subtorus have zero last two coordinates. They remain independent in the first two coordinates, so the fixed point in $\mathbb{G}_m^2$ lies in a finite subgroup and is torsion. At the specified point, both ξ^m and ξ^n are roots of unity, forcing ξ to be a root of unity. This is a contradiction. The other constant projection is symmetric.

Both u and v are therefore nonconstant. The annihilator lattice of the two-dimensional subtorus has rank two. Its two independent characters lie in the kernel of the divisor map

$$\mathbb{Q}^4 \longrightarrow \text{Div}(C) \otimes \mathbb{Q}$$

sending the standard basis to $\text{div}(u), \text{div}(u-1), \text{div}(v), \text{div}(v-1)$. Hence the four divisors span a space of dimension at most two. Lemma 17.2 shows that $v = \sigma(u)$ for one of its six Möbius transformations.

At the point coming from ξ , this gives two binomial equations

$$\xi^{p-r_\sigma} = c_1, \quad \xi^{q-s_\sigma} = c_2, \quad c_1, c_2 \in \mathbb{Q}^\times,$$

where

$\sigma(u)$	r_σ	s_σ
u	m	n
$1-u$	n	m
u^{-1}	$-m$	$n-m$
$(1-u)^{-1}$	$-n$	$m-n$
$u/(u-1)$	$m-n$	$-n$
$(u-1)/u$	$n-m$	$-m$

The two exponent differences cannot both vanish unless the ordered edges, and hence the packages, agree. Put

$$d = \gcd(|p - r_\sigma|, |q - s_\sigma|) > 0.$$

Bézout gives $\xi^d = c \in \mathbb{Q}^\times$. If a prime divided both d and $\gcd(m, n)$, it would divide r_σ, s_σ, p, q , contrary to the coprimality hypothesis. Thus

$$\gcd(d, \gcd(m, n)) = 1.$$

The minimal polynomial of ξ divides both $H_{m,n}$ and $x^d - c$. Lemma 14.1 gives degree at most one, and Theorem 10.1 then permits only the two orientations of the single package $\{1, 2\}$. Two distinct packages are impossible. $\square$

We now state the sparse-gcd handoff in precisely the form used below. For a torus homomorphism ψ , write $\psi^\#$ for pullback on Laurent polynomials.

Proposition 17.4 (Specialized Amoroso–Sombra–Zannier output). *Let*

$$\varphi(t) = (t^m, t^n, t^p, t^q), \quad D = \max\{m, n, p, q\},$$

where $\gcd(m, n, p, q) = 1$, and let $\xi \notin \mu_\infty$ be a common root of $H_{m,n}$ and $H_{p,q}$. Put

$$L_1 = (n - m) - nx_1 + mx_2, \quad L_2 = (q - p) - qx_3 + px_4.$$

Let $B_4 = B'(4, 1)$ be the effective constant in Theorem 2.6 of [2]. If

$$D^{1/6} > B_4(1 + \log(2D)), \tag{17.3}$$

then there are $0 \leq k \leq 3$, an injective torus homomorphism

$$\psi : \mathbb{G}_m^{4-k} \longrightarrow \mathbb{G}_m^4$$

of size at most B_4 , and $\varphi_1 : \mathbb{G}_m \rightarrow \mathbb{G}_m^{4-k}$, such that $\varphi = \psi \circ \varphi_1$. Put

$$F_i = \psi^\# L_i, \quad y_0 = \varphi_1(\xi).$$

Either both F_i vanish identically, or, with

$$G = \gcd(F_1, F_2), \quad g = \varphi_1^\# G,$$

one has $G(y_0) = 0$.

Proof. The coefficient height is at most $\log(2D)$, and the exponent vector is primitive, so Theorem 2.6 applies under (17.3). It gives the displayed factorization and says that $g \mid \gcd(H_{m,n}, H_{p,q})$. Moreover, a root of the quotient of this gcd by g is either torsion or is a common root of the two proper-subsum equations obtained by retaining a nonempty proper subset $\Lambda \subset \{1, 2, 3, 4\}$ of the four nonconstant monomials, while retaining the constant terms.

The point ξ is not torsion. Let $I = \Lambda \cap \{1, 2\}$. If I is empty, the first partial equation is the nonzero constant $n - m$. If $I = \{1\}$, it is $(n - m) - n\xi^m = 0$; subtracting it from the full first equation gives $m\xi^n = 0$, impossible. If $I = \{2\}$, subtraction gives $-n\xi^m = 0$, also impossible. Thus $I = \{1, 2\}$. The second equation similarly forces $\Lambda \cap \{3, 4\} = \{3, 4\}$. Hence Λ would be the full set, contradicting properness. Therefore $g(\xi) = 0$, which is exactly $G(y_0) = 0$. The possibility that both pulled-back forms vanish identically is recorded separately. $\square$

Theorem 17.5 (Effective absolute boundedness of normalized counterexamples). *There is an effectively computable absolute constant D_0 such that a normalized four-index counterexample*

$$0 < m < n, \quad 0 < p < q, \quad \gcd(m, n, p, q) = 1$$

must satisfy

$$\max\{m, n, p, q\} \leq D_0.$$

More explicitly, with B_4 as in Proposition 17.4, define

$$D_{\text{ASZ}} = \max\left(\{1\} \cup \left\{D \in \mathbb{N} : D^{1/6} \leq B_4(1 + \log(2D))\right\}\right), \quad D_0 = \max\{\lceil B_4 \rceil, D_{\text{ASZ}}\}. \quad (17.4)$$

Then this D_0 has the stated property.

Proof. Assume $D > D_0$ and use Proposition 17.4. The two pulled-back forms cannot both vanish identically. Indeed, if

$$(n - m) - n\chi_1 + m\chi_2 = 0$$

as a Laurent polynomial in characters of a positive-dimensional torus, linear independence of distinct characters forces $\chi_1 = \chi_2 = 1$. The second form similarly forces $\chi_3 = \chi_4 = 1$, making ψ trivial, contrary to injectivity. Thus $G(y_0) = 0$. Choose a geometric irreducible factor P of G vanishing at y_0 ; if exactly one F_i is zero, take P from the other one.

If $k = 3$, write the primitive exponent column of ψ as $B = (b_1, b_2, b_3, b_4)^\top$, so

$$(m, n, p, q)^\top = B\theta.$$

Injectivity gives $\gcd(b_1, b_2, b_3, b_4) = 1$, while normalization gives $\gcd(m, n, p, q) = 1$. Hence $|\theta| = 1$ and $D = \|B\|_\infty \leq B_4$, contradicting $D > D_0$.

If $k = 0$, the injective endomorphism ψ of a torus of the same dimension is an automorphism. Pullback therefore preserves coprimality, whereas L_1, L_2 are coprime Laurent linear forms. Thus G is a unit, contradicting $G(y_0) = 0$.

Suppose $k = 1$. The hypersurface $V(P) \subset \mathbb{G}_m^3$ is a geometric irreducible surface. An injective homomorphism of tori is a closed immersion, so $\psi(V(P))$ is a closed irreducible surface contained in

$$X = V(L_1, L_2),$$

the irreducible product of two punctured lines. Since $\dim X = 2$, the image must equal X , placing X inside the proper subtorus $\text{im } \psi$.

This is impossible even for a mixed monomial relation. If

$$x_1^a x_2^b x_3^c x_4^d$$

were constant on the product, fixing the second factor would make $x_1^a x_2^b$ constant on the first line, and fixing the first would do the same for $x_3^c x_4^d$ on the second. Parameterizing the first line by

$$x_1 = t, \quad x_2 = \frac{nt - (n - m)}{m}$$

and comparing the valuations at $0, (n - m)/n, \infty$ gives $a = b = 0$; symmetrically $c = d = 0$. Thus X lies in no proper subtorus.

Finally let $k = 2$. The geometric curve $V(P) \subset \mathbb{G}_m^2$ has closed image contained in $X \cap \text{im } \psi$ and passing through the specified point. The global normalization gives

$$\gcd(\gcd(m, n), \gcd(p, q)) = 1.$$

Proposition 17.3 excludes this curve. Every possible value of k is contradictory, proving the bound. $\square$

Corollary 17.6 (Finite-template reduction). *Every counterexample is a simultaneous positive integral dilation of one of finitely many normalized configurations. In particular, the full package coprimality conjecture is reduced to an effective finite exact computation.*

Proof. Divide the four edge endpoints by their common gcd and replace the common root z by the corresponding power, as in (2.6). The normalized configuration is bounded by Theorem 17.5. Conversely, the original configuration is its simultaneous dilation. $\square$

Remark 17.7 (The symbolic ASZ threshold). The constant B_4 is effective in the cited theorem, but no numerical value is supplied there. Thus Theorem 17.5, by itself, does not furnish a practical endpoint. The direct argument in the next subsection bypasses B_4 and produces an explicit numerical bound.

17.1 A direct numerical bound from height, geometry of numbers, and Smyth's theorem

The preceding Amoroso–Sombra–Zannier argument proves effective absolute boundedness without displaying a numerical constant. In the present four-variable situation one can obtain a substantially more concrete bound by combining an elementary height estimate with a short rank-two subtorus. We write $h(\alpha)$ for the absolute logarithmic Weil height and $M(f)$ for the Mahler measure of a primitive polynomial $f \in \mathbb{Z}[x]$.

Lemma 17.8 (Height of a collision root). *Let $0 < u < v$, and suppose $H_{u,v}(\alpha) = 0$. Then*

$$(v - u) h(\alpha) \leq \log(2v), \quad u h(\alpha) \leq \log(2v). \quad (17.5)$$

Consequently,

$$v h(\alpha) \leq 2 \log(2v). \quad (17.6)$$

Proof. We use the elementary affine-height inequality

$$h\left(\frac{a\beta + b}{c}\right) \leq h(\beta) + \log(2 \max\{|a|, |b|, |c|\}), \quad a, b, c \in \mathbb{Z}, \quad c \neq 0.$$

It follows place by place from the triangle inequality, with the factor two needed only at the archimedean places. The collision equation gives

$$\alpha^v = \frac{v\alpha^u - (v - u)}{u},$$

and hence

$$v h(\alpha) \leq u h(\alpha) + \log(2v),$$

which is the first inequality in (17.5).

The reciprocal identity

$$x^v H_{v-u,v}(x^{-1}) = H_{u,v}(x)$$

shows that α^{-1} is a root of $H_{v-u,v}$. Applying the first inequality to that edge and using $h(\alpha^{-1}) = h(\alpha)$ gives the second inequality. Adding them proves (17.6). $\square$

For a Euclidean lattice Λ , write $\det \Lambda$ for its covolume in its real span.

Lemma 17.9 (Degree of a rank-two subtorus). *Let $R \subset \mathbb{Z}^4$ be a saturated rank-two lattice, and let $T \subset \mathbb{G}_m^4$ be the two-dimensional subtorus annihilated by R . Under the standard embedding*

$$\mathbb{G}_m^4 \longrightarrow \mathbb{P}^4, \quad (x_1, x_2, x_3, x_4) \longmapsto [1 : x_1 : x_2 : x_3 : x_4],$$

the projective closure $\overline{T}$ satisfies

$$\deg \overline{T} \leq 2 \det R. \tag{17.7}$$

Proof. Put $M = \mathbb{Z}^4/R$. Saturation makes M a free rank-two lattice. Choose a lattice basis of M , and let $u_1, \dots, u_4 \in \mathbb{Z}^2$ be the images of the four standard basis vectors. With

$$\mathcal{A} = \{0, u_1, u_2, u_3, u_4\}, \quad P = \text{conv}(\mathcal{A}),$$

the closure $\overline{T}$ is the projective toric surface $X_{\mathcal{A}}$. The vectors u_i generate M , so there is no lattice-index correction. The normalization of $X_{\mathcal{A}}$ is the normal toric surface associated with P , and finite birational normalization preserves the hyperplane degree. Hence the standard degree–volume theorem gives

$$\deg \overline{T} = \text{Vol}_M(P),$$

where Vol_M is normalized lattice area: it is twice ordinary Euclidean area in coordinates in which a fundamental parallelogram of M has area one [6, Theorems 3.A.5 and 13.4.3].

Let U be the 2×4 integer matrix with columns u_i , and let $S \subset \mathbb{Z}^4$ be its row lattice. The quotient map $U : \mathbb{Z}^4 \rightarrow \mathbb{Z}^2$ is surjective and has kernel R . Thus S is primitive; since its real span is $R^\perp$,

$$S = R^\perp \cap \mathbb{Z}^4.$$

For complementary primitive rank-two lattices in the unimodular lattice $\mathbb{Z}^4$, the covolumes agree. Concretely, the Euclidean Hodge star sends a primitive generator of $\wedge^2 R$ to a primitive generator

of $\wedge^2 S$ and preserves norm. Hence $\det S = \det R$. The two rows of U form a basis of S , so Cauchy–Binet gives

$$(\det R)^2 = (\det S)^2 = \det(UU^\top) = \sum_{1 \leq i < j \leq 4} \det(u_i, u_j)^2. \quad (17.8)$$

Order the vertices of P cyclically. In the shoelace sum, every edge incident with the vertex 0 contributes zero. All other vertices belong to $\{u_1, \dots, u_4\}$, so at most four nonzero determinant terms remain. Therefore Cauchy–Schwarz and (17.8) give

$$\deg \bar{T} \leq 2 \left(\sum_{i < j} \det(u_i, u_j)^2 \right)^{1/2} = 2 \det R.$$

□

Proposition 17.10 (A short rank-two subtorus). *Let*

$$\mathbf{a} = (m, n, p, q) \in \mathbb{Z}^4$$

be primitive, and put $D = \|\mathbf{a}\|_\infty$. There is a two-dimensional subtorus $T \subset \mathbb{G}_m^4$ containing the one-parameter subgroup

$$t \mapsto (t^m, t^n, t^p, t^q)$$

and satisfying

$$\deg \bar{T} < 4D^{2/3}. \quad (17.9)$$

Proof. Let

$$L = \mathbf{a}^\perp \cap \mathbb{Z}^4.$$

Primitivity of $\mathbf{a}$ means that the homomorphism $z \mapsto \langle \mathbf{a}, z \rangle$ maps $\mathbb{Z}^4$ onto $\mathbb{Z}$. Thus L is saturated of rank three. Choose $c \in \mathbb{Z}^4$ with $\langle \mathbf{a}, c \rangle = 1$. A basis of L , together with c , is a basis of $\mathbb{Z}^4$; comparing base volume with the height $1/\|\mathbf{a}\|_2$ above $\mathbf{a}^\perp$ gives

$$\det L = \|\mathbf{a}\|_2 \leq 2D.$$

Its Euclidean dual lattice $L^\vee$ has determinant $(\det L)^{-1}$. Minkowski’s first theorem for the three-dimensional Euclidean ball [5, Chapter III, §2] provides a nonzero $y \in L^\vee$ with

$$\|y\|_2 \leq \left(\frac{6}{\pi} \right)^{1/3} (\det L)^{-1/3}.$$

Divide by the content of y in the free abelian group $L^\vee$. This can only decrease its norm, and it makes y primitive, meaning that

$$L \longrightarrow \mathbb{Z}, \quad b \mapsto \langle b, y \rangle$$

is surjective. Set

$$R = \{b \in L : \langle b, y \rangle = 0\}.$$

Then $L/R \simeq \mathbb{Z}$, so R is saturated in L . It is also saturated in $\mathbb{Z}^4$: in the exact sequence

$$0 \longrightarrow L/R \longrightarrow \mathbb{Z}^4/R \longrightarrow \mathbb{Z}^4/L \longrightarrow 0$$

both outer groups are torsion-free, hence so is $\mathbb{Z}^4/R$.

Choose a basis b_1, b_2 of R and $b_3 \in L$ with $\langle b_3, y \rangle = 1$. The Euclidean height of b_3 above $R \otimes_{\mathbb{Z}} \mathbb{R}$ is $1/\|y\|_2$. Therefore

$$\det L = \frac{\det R}{\|y\|_2}, \quad \det R = (\det L) \|y\|_2 \leq \left(\frac{6}{\pi}\right)^{1/3} (2D)^{2/3}.$$

Let T be the subtorus annihilated by R . Since $R \subset \mathbf{a}^\perp$, every character in R is trivial on the displayed one-parameter subgroup; hence T contains that subgroup. Lemma 17.9 gives

$$\deg \bar{T} \leq 2 \left(\frac{6}{\pi}\right)^{1/3} (2D)^{2/3} = 2 \left(\frac{24}{\pi}\right)^{1/3} D^{2/3} < 4D^{2/3},$$

because $\pi > 3$. □

Proposition 17.11 (Degree of a normalized common-root field). *Let*

$$0 < m < n, \quad 0 < p < q, \quad \gcd(m, n, p, q) = 1,$$

let the two packages be distinct, and suppose that $\alpha \notin \mu_\infty$ is a common root of $H_{m,n}$ and $H_{p,q}$. Put

$$D = \max\{m, n, p, q\}, \quad d = [\mathbb{Q}(\alpha) : \mathbb{Q}].$$

Then

$$d < 4D^{2/3}. \tag{17.10}$$

Proof. Take the subtorus T from Proposition 17.10. It contains

$$P_\alpha = (\alpha^m, \alpha^n, \alpha^p, \alpha^q).$$

The variety cut out in $\mathbb{G}_m^4$ by the two collision lines is the variety X of Proposition 17.3. That proposition shows that P_α is isolated in $T \cap X$: a positive-dimensional local component would contain a geometric irreducible curve through the point. Since P_α lies in the open torus, a positive-dimensional component of $\bar{T} \cap \bar{X}$ through it would restrict near that point to one in $T \cap X$. Hence it is also an isolated projective intersection point.

Both T and X , and therefore their projective closures, are defined over $\mathbb{Q}$. The closure $\bar{X}$ is the intersection of the two rational hyperplanes

$$(n-m)x_0 - nx_1 + mx_2 = 0, \quad (q-p)x_0 - qx_3 + px_4 = 0$$

in $\mathbb{P}^4$, so $\deg \bar{X} = 1$. The generalized Bezout inequality for isolated components [7, §8.4] gives

$$\deg((\bar{T} \cap \bar{X})_{\text{isol}}) \leq \deg \bar{T} \deg \bar{X} = \deg \bar{T}.$$

Every Galois conjugate of P_α is again isolated and contributes positive intersection multiplicity. The closed point determined by this orbit therefore contributes at least $[\mathbb{Q}(P_\alpha) : \mathbb{Q}]$, whence

$$[\mathbb{Q}(P_\alpha) : \mathbb{Q}] \leq \deg \bar{T} < 4D^{2/3}.$$

Finally, choose integers r_1, r_2, r_3, r_4 with $r_1 m + r_2 n + r_3 p + r_4 q = 1$. Since all four coordinates are nonzero,

$$\alpha = (\alpha^m)^{r_1} (\alpha^n)^{r_2} (\alpha^p)^{r_3} (\alpha^q)^{r_4}.$$

Thus $\mathbb{Q}(P_\alpha) = \mathbb{Q}(\alpha)$, proving (17.10). □

Let

$$\theta_0 = 1.3247179572447460259 \dots$$

be the plastic number, the real root greater than one of $T^3 - T - 1$.

Proposition 17.12 (A uniform Mahler-measure gap). *Let $h \in \mathbb{Z}[x]$ be the primitive irreducible polynomial, with positive leading coefficient, of a noncyclotomic common root of two collision trinomials. Then h is nonreciprocal and*

$$M(h) \geq \theta_0. \quad (17.11)$$

Proof. The polynomial h divides an orientation $Q_{a,b}$. If it were self-reciprocal, it would also divide the reciprocal orientation $Q_{b,a}$, contradicting the nonzero reciprocal resultant (16.3). Thus h is nonreciprocal.

If h is monic, Smyth's theorem [12] gives $M(h) \geq \theta_0$. If it is not monic, its positive integral leading coefficient is at least two, and $M(h) \geq 2 > \theta_0$. $\square$

Theorem 17.13 (Explicit absolute bound for normalized counterexamples). *Every normalized four-index counterexample*

$$0 < m < n, \quad 0 < p < q, \quad \gcd(m, n, p, q) = 1$$

satisfies

$$\boxed{\max\{m, n, p, q\} \leq 175\,394\,637.} \quad (17.12)$$

Proof. Let α be a common root, let h be its primitive irreducible polynomial, let $d = \deg h$, and put $D = \max\{m, n, p, q\}$. One of the two collision edges has upper endpoint D . Lemma 17.8 gives

$$h(\alpha) \leq \frac{2 \log(2D)}{D}. \quad (17.13)$$

By Proposition 17.11, $d < 4D^{2/3}$. Since $\log M(h) = d h(\alpha)$, Proposition 17.12 yields

$$\log \theta_0 \leq \log M(h) < \frac{8 \log(2D)}{D^{1/3}}. \quad (17.14)$$

The function on the right is strictly decreasing for $D \geq 11$. Certified real-ball arithmetic gives

$$\frac{8 \log(2 \cdot 175394637)}{175394637^{1/3}} > \log \theta_0,$$

whereas

$$\frac{8 \log(2 \cdot 175394638)}{175394638^{1/3}} < \log \theta_0.$$

Thus (17.14) is impossible for $D \geq 175394638$. The smaller values $D < 11$ are already below the asserted bound. $\square$

Corollary 17.14 (The upper range is unit-normalized). *In a normalized counterexample with*

$$D = \max\{m, n, p, q\} \geq 6\,816\,242,$$

the primitive irreducible polynomial h of a common root is monic and has constant coefficient ± 1 .

Proof. The same decreasing function satisfies

$$\frac{8 \log(2 \cdot 6816242)}{6816242^{1/3}} < \log 2.$$

Hence (17.13) and Proposition 17.11 give $M(h) < 2$. Both the absolute leading coefficient and the absolute constant coefficient of a primitive polynomial are at most its Mahler measure. They are nonzero integers, so both equal one. $\square$

Remark 17.15. Theorem 17.13 removes the practical indeterminacy in the constant B_4 . The Amoroso–Sombra–Zannier route remains an independent structural proof of effective boundedness, while the direct argument gives a fully numerical endpoint. The two tiny numerical comparisons above can be certified by the real-ball script recorded in Section 18.

17.2 Moment, discriminant, and endpoint sieves

The explicit box is still large. The following additional restrictions are intended for the finite computation. They are extracted from the all-moment form of Borisov’s cluster-count argument.

Let

$$h(x) = ux^e + c_{e-1}x^{e-1} + \cdots + c_1x + v \in \mathbb{Z}[x], \quad u > 0,$$

be primitive and irreducible. For its roots β , counted with multiplicity, put

$$A_j(h) = \sum_{h(\beta)=0} (1 - \beta)\beta^j \quad (j \geq 0). \quad (17.15)$$

Proposition 17.16 (All-moment congruence). *Suppose h divides two selected primitive orientations $Q_{A,B}$ and $Q_{C,D}$, where*

$$(A, B) = (C, D) = 1, \quad N = A + B, \quad M = C + D.$$

If a prime ℓ divides $ABNCDM$ but does not divide uv , then

$$A_j(h) \in \ell\mathbb{Z}_{(\ell)} \quad \text{for every } j \geq 0. \quad (17.16)$$

Proof. Use a package whose additive triple contains ℓ . Since $\ell \nmid uv$, the leading and constant coefficients of h are ℓ -adic units. The Newton polygon of h is therefore horizontal, so all roots of h are ℓ -adic units.

Borisov’s cluster argument [1, Lemma 3.1 and Remark 3.5] shows that, at every nontrivial prime-to- ℓ root-of-unity center, the number of roots belonging to a rational factor is divisible by ℓ . Modulo ℓ , the summand in (17.15) is constant on such a cluster, namely $(1 - \zeta)\zeta^j$, so the cluster contribution vanishes. At the center 1, every summand already vanishes modulo ℓ . If ℓ is an endpoint prime, the nonunit endpoint block is absent because every root of h is a unit. These contributions exhaust the roots and prove (17.16). The argument also covers $\ell = 2$; the merged side still contributes an even number of roots to every rational factor, as in Proposition 15.4. $\square$

Define the reciprocal coefficient polynomial

$$\tilde{h}(t) = t^e h(t^{-1}) = u + c_{e-1}t + \cdots + c_1t^{e-1} + vt^e.$$

Corollary 17.17 (Differential moment congruence). *Under the hypotheses of Proposition 17.16, put*

$$\mathcal{S} = \text{rad}(ABNC DM), \quad \mathcal{R} = \prod_{\substack{\ell | \mathcal{S} \\ \ell \nmid uv}} \ell.$$

Then every coefficient of

$$(1-t)\tilde{h}'(t) + e\tilde{h}(t) \tag{17.17}$$

is divisible by $\mathcal{R}$. In particular,

$$\mathcal{R} \mid eu + c_{e-1}, \quad \mathcal{R} \mid ev + c_1. \tag{17.18}$$

Proof. For each prime in $\mathcal{R}$, the constant coefficient $\tilde{h}(0) = u$ is a unit. The generating series of the moments is

$$\sum_{j \geq 0} A_j(h) t^j = e + (1-t) \frac{\tilde{h}'(t)}{\tilde{h}(t)}.$$

Proposition 17.16 makes the left side zero modulo the prime. Multiplication by $\tilde{h}(t)$ gives (17.17), and the constant and t^{e-1} coefficients give (17.18). $\square$

Proposition 17.18 (Hankel–discriminant divisibility). *With the same notation,*

$$\boxed{\mathcal{R}^e \mid h(1) \text{Disc}(h)}. \tag{17.19}$$

Consequently,

$$\mathcal{R} \leq 2eM(h)^{2-1/e}. \tag{17.20}$$

Proof. Form the $e \times e$ Hankel matrix

$$\mathcal{H} = (A_{i+j}(h))_{0 \leq i, j < e}.$$

Writing $w_k = 1 - \beta_k$, its Vandermonde factorization is

$$\mathcal{H} = V \text{diag}(w_1, \dots, w_e) V^T.$$

Hence, up to sign,

$$\det \mathcal{H} = \prod_k (1 - \beta_k) \prod_{i < j} (\beta_i - \beta_j)^2 = \frac{h(1) \text{Disc}(h)}{u^{2e-1}}. \tag{17.21}$$

Every entry is divisible by every prime in $\mathcal{R}$, while u is a unit at those primes. Thus (17.19) follows.

The elementary estimates

$$|h(1)| \leq 2^e M(h), \quad |\text{Disc}(h)| \leq e^e M(h)^{2e-2}$$

(the second is Mahler’s discriminant inequality [11]) imply

$$\mathcal{R}^e \leq 2^e e^e M(h)^{2e-1}.$$

Taking e -th roots proves (17.20). $\square$

Corollary 17.19 (Support bound for two colliding primitive packages). *Under the same hypotheses,*

$$\mathcal{S} \leq 2eM(h)^{4-1/e}. \quad (17.22)$$

If h is monic with constant coefficient ± 1 , then

$$\mathcal{S} \leq 2eM(h)^{2-1/e}. \quad (17.23)$$

Proof. The primes of $\mathcal{S}$ omitted from $\mathcal{R}$ divide uv . Since

$$\text{rad}(uv) \leq |uv| \leq M(h)^2,$$

formula (17.20) gives (17.22). In the unit-normalized case no prime is omitted, so $\mathcal{S} = \mathcal{R}$, giving (17.23). $\square$

Proposition 17.20 (Endpoint divisibilities and slope matching). *Under the same hypotheses,*

$$|v|^A \mid B^e, \quad u^B \mid A^e, \quad |v|^C \mid D^e, \quad u^D \mid C^e. \quad (17.24)$$

Moreover, for every prime $\ell \mid u$,

$$\frac{v_\ell(A)}{B} = \frac{v_\ell(C)}{D}, \quad (17.25)$$

and, for every prime $\ell \mid v$,

$$\frac{v_\ell(B)}{A} = \frac{v_\ell(D)}{C}. \quad (17.26)$$

Proof. Because $Q_{A,B}$ is primitive with leading coefficient A and constant coefficient B , Gauss's lemma gives an integral complementary factor. Hence $u \mid A$, $|v| \mid B$, and $(u, v) = 1$. At a root β of h ,

$$\beta^A(N - A\beta^B) = B.$$

Taking the resultant with h gives

$$\text{Res}(h, N - Ax^B) = \pm \frac{B^e u^N}{v^A} \in \mathbb{Z}. \quad (17.27)$$

Since $(u, v) = 1$, this proves $|v|^A \mid B^e$. Applying the same argument to the reciprocal factor in $Q_{B,A}$ proves $u^B \mid A^e$. The second package gives the remaining two divisibilities.

Now let $\ell \mid u$. Then $\ell \mid A, C$ and $\ell \nmid v$. The product of the roots of h has negative ℓ -adic valuation, so at least one root is nonunit. Borisov's endpoint polygons say that every negative-valuation root of $Q_{A,B}$ has valuation $-v_\ell(A)/B$, and every such root of $Q_{C,D}$ has valuation $-v_\ell(C)/D$. The same root lies in both polynomials, proving (17.25). The positive-valuation endpoint description gives (17.26) in exactly the same way when $\ell \mid v$. $\square$

Remark 17.21. For a finite search, the support bound, the four endpoint divisibilities, and the slope equalities should be applied before any polynomial factorization. For a surviving exponent tuple, let $S \in \mathbb{Z}[x]$ be the primitive part of $W_{m,n;p,q}/(x-1)^4$. Proposition 17.1 then permits a cheap modular certificate: if an auxiliary prime does not divide the leading coefficient of S —equivalently, reduction preserves its degree—and $\bar{S}$ is square-free over the corresponding finite field, then S is square-free over $\mathbb{Q}$, so no common irreducible factor can exist. The good-reduction condition is essential: at a prime where the degree drops, a square factor can disappear after reduction.

18 Exact upper-range elimination and reproducibility

Put

$$D_0 = 6\,816\,242, \quad D_1 = 175\,394\,637.$$

This section gives the mathematical reductions and the exact computation which exclude every normalized counterexample with $D_0 \leq D \leq D_1$. Write the two unordered packages in primitive form as

$$\{rA, rB\}, \quad \{sC, sE\}, \quad \gcd(A, B) = \gcd(C, E) = 1,$$

put $N = A + B$, $M = C + E$, and order the shapes by $A < B$, $C < E$. The corresponding collision upper endpoints are rN and sM , so

$$D = \max(rN, sM).$$

Normalization gives $\gcd(r, s) = 1$. Let α be a common root, let h_α be its primitive minimal polynomial, and put

$$d = [\mathbb{Q}(\alpha) : \mathbb{Q}], \quad \beta = \alpha^r, \quad \gamma = \alpha^s.$$

The elements β and γ are roots of selected reciprocal orientations of the two primitive shapes.

18.1 An exact uniform degree gap and small scales

Lemma 18.1 (An algebraic certificate for the plastic number). *Let θ_0 be the real root greater than one of $T^3 - T - 1$. Then*

$$\theta_0^{70} = 62\,608\,681 + 109\,870\,576\theta_0 + 82\,938\,844\theta_0^2 > 2D_1. \quad (18.1)$$

Proof. Reduction of T^{70} modulo $T^3 - T - 1$ gives the displayed identity. The polynomial $T^3 - T - 1$ is strictly increasing for $T \geq 1$, and

$$\left(\frac{33}{25}\right)^3 - \frac{33}{25} - 1 = -\frac{313}{15625} < 0.$$

Hence $\theta_0 > 33/25$. Substitution in the positive-coefficient right-hand side of (18.1) gives the exact rational lower bound

$$\theta_0^{70} > \frac{220\,094\,051\,941}{625} > 350\,789\,274 = 2D_1.$$

□

Proposition 18.2 (Uniform degree and scale bounds). *For a normalized counterexample with $D \leq D_1$,*

$$d > \frac{D}{140}. \quad (18.2)$$

In the primitive decomposition above,

$$1 \leq r, s \leq 139. \quad (18.3)$$

In particular, throughout the unit range $D \geq D_0$,

$$d \geq \left\lfloor \frac{D_0}{140} \right\rfloor + 1 = 48\,688.$$

Proof. The height estimate (17.13) and the Mahler gap (17.11) give

$$d \geq \frac{D \log \theta_0}{2 \log(2D)}.$$

Lemma 18.1 gives $70 \log \theta_0 > \log(2D_1) \geq \log(2D)$, proving (18.2). The confluent degree bound, Theorem 14.2, gives

$$d \leq \min(N, M) - 2 < N, M.$$

Since $rN \leq D$ and $sM \leq D$, the inequalities $D/140 < d < N \leq D/r$ and $D/140 < d < M \leq D/s$ imply $r, s < 140$. $\square$

18.2 Full Kummer degree in the unit range

The prime-degree form of Capelli's criterion used earlier has the following standard composite-degree form: over a characteristic-zero field K , $X^n - a$ is irreducible if $a \notin K^p$ for every prime $p \mid n$, and, when $4 \mid n$, also $a \notin -4K^4$; see [10, Chapter VI, Section 9].

Proposition 18.3 (Full Kummer degree). *Assume $D_0 \leq D \leq D_1$. Let $e_1 = [\mathbb{Q}(\beta) : \mathbb{Q}]$ and $e_2 = [\mathbb{Q}(\gamma) : \mathbb{Q}]$. Then*

$$X^r - \beta \text{ is irreducible over } \mathbb{Q}(\beta), \quad X^s - \gamma \text{ is irreducible over } \mathbb{Q}(\gamma).$$

Consequently there is an integer $k \geq 1$ such that

$$d = re_1 = se_2 = rsk, \quad e_1 = sk, \quad e_2 = rk. \quad (18.4)$$

Proof. Corollary 17.14 makes α , and hence β, γ , algebraic units. Fix a prime $p \mid r$, put $K = \mathbb{Q}(\beta)$, and suppose first that $\beta \in K^p$.

If $p \mid N$, Corollary 15.5 gives an immediate contradiction. If $p \mid AB$, the proof of Theorem 15.6 shows that an internal p -th power factor must lie entirely in the nonunit endpoint block, again impossible for the unit β . The only remaining case is $p \nmid ABN$. Then p is odd, and Theorem 14.5 gives $e_1 \leq p - 2$. Since α satisfies $X^r - \beta$ over K ,

$$d \leq re_1 \leq 139 \cdot 137 = 19\,043,$$

contrary to $d \geq 48\,688$. Thus $\beta \notin K^p$ for every prime $p \mid r$.

It remains to exclude Capelli's exceptional fourth-power case. If $4 \mid r$, then 2 occupies exactly one role in the primitive triple (A, B, N) . The local proofs of Corollary 15.5 and Theorem 15.6, including the residual splitting of the merged side, place every unit root, after an unramified base extension, in a packet for which

$$v_2(\beta - \zeta) = \frac{1}{\ell} < 1, \quad [L(\beta) : L] \leq \ell,$$

with ζ a root of unity of odd order and $\ell \geq 2$. Since

$$-\beta - \zeta = -((\beta - \zeta) + 2\zeta),$$

the two summands have unequal valuations and $v_2(-\beta - \zeta) = 1/\ell$. Lemma 14.3 applied to $-\beta$ shows that $-\beta$ is not a square in K . In particular $\beta \notin -4K^4$, because $\beta = -4\eta^4$ would give $-\beta = (2\eta^2)^2$. Capelli's criterion now proves the irreducibility of $X^r - \beta$. The proof for $X^s - \gamma$ is identical.

It follows that $d = re_1 = se_2$. Since $\gcd(r, s) = 1$, one has $s \mid e_1$ and $r \mid e_2$, giving (18.4). $\square$

18.3 One-package support and full-radical congruences

Proposition 18.4 (One-package unit support bound). *Let $f \in \mathbb{Z}[x]$ be a monic irreducible factor of a selected primitive orientation of shape (A, B) , let $e = \deg f$, and suppose $f(0) = \pm 1$. Then*

$$\text{rad}(AB(A+B)) \leq 2eM(f)^{2-1/e}. \quad (18.5)$$

Proof. For every prime dividing $AB(A+B)$, the proof of Proposition 17.16 uses only the one primitive package containing that prime. Since the leading and constant coefficients of f are units, its endpoint nonunit block is absent. The same all-moment congruence, Hankel determinant, and Mahler discriminant estimate as in Proposition 17.18 therefore apply with $\mathcal{R} = \text{rad}(AB(A+B))$, giving (18.5). $\square$

Corollary 18.5 (The exact support cap used by the computation). *Under the hypotheses of Proposition 18.3,*

$$r \text{rad}(ABN) < 4\,398\,937, \quad s \text{rad}(CEM) < 4\,398\,937. \quad (18.6)$$

Proof. Let f_β be the minimal polynomial of β . From (18.4),

$$\log M(f_\beta) = e_1 h(\beta) = e_1 r h(\alpha) = d h(\alpha) = \log M(h_\alpha).$$

Since β is noncyclotomic, $M(f_\beta) > 1$. By Propositions 17.11 and 18.4, and using $e_1 < 4D^{2/3}/r$ together with (17.13), we obtain

$$\begin{aligned} r \text{rad}(ABN) &\leq 2re_1 M(f_\beta)^{2-1/e_1} \\ &< 8D^{2/3} \exp\left(\frac{16 \log(2D)}{D^{1/3}}\right) =: C(D). \end{aligned}$$

Indeed, $\log M(f_\beta) = \log M(h_\alpha)$ and $\log M(h_\alpha) < 8 \log(2D)/D^{1/3}$. Writing $x = D^{1/3}$,

$$\frac{d}{dx} \log C(x^3) = \frac{2x + 48 - 16 \log(2x^3)}{x^2} > 0$$

throughout $D_0 \leq D \leq D_1$: here $x > 189$ and $\log(2D) < 20$, so the numerator is greater than 106. Thus it is enough to bound $C(D_1)$. The exact rational certificate in `code/certify_constants.py` uses

$$\frac{559\,764\,608}{10^6} < D_1^{1/3} < \frac{559\,764\,609}{10^6}$$

and rational upper bounds from the atanh series for $\log$ and the Taylor series for $\exp$. It proves

$$C(D_1) < 4\,398\,937.$$

The second package is symmetric. $\square$

Proposition 18.6 (Full-radical degree signature in the unit case). *Under the hypotheses of Proposition 18.4, for every prime $p \mid AB(A+B)$,*

$$e \equiv 0 \quad \text{or} \quad -2 \pmod{p}. \quad (18.7)$$

Thus $\text{rad}(AB(A+B)) \mid e(e+2)$.

Proof. For a total prime this is Theorem 15.1. At an endpoint prime the unit hypothesis removes the endpoint nonunit block. Borisov's remaining root-of-unity clusters have contributions divisible by p , except that the center-1 packet contributes either 0 or $p-2$ modulo odd p . For $p=2$, every packet contribution is even, including the residual-split merged side. This proves (18.7) in every role. $\square$

18.4 Role-binomial degree bounds

Lemma 18.7 (Reduction to a role binomial). *Let p divide exactly one member R of the primitive triple (A, B, N) , and let h_α be monic with constant coefficient ± 1 . If h_α divides a selected orientation $Q_{A,B}(X^r)$ or $Q_{B,A}(X^r)$, then*

$$\overline{h_\alpha} \mid X^{rR} - 1 \quad \text{in } \mathbb{F}_p[X], \quad d \leq rR. \quad (18.8)$$

If the same prime occupies role S in the second primitive triple, then

$$d \leq \gcd(rR, sS).$$

Proof. For either orientation, direct reduction of the collision trinomial gives

$$H_{T,N}(X) \equiv cX^\nu(X^R - 1) \pmod{p}$$

for some $c \in \mathbb{F}_p^\times$, $\nu \geq 0$, where $T \in \{A, B\}$ is the selected endpoint. Hence

$$(X^r - 1)^2 Q_{T,N-T}(X^r) = H_{T,N}(X^r) \equiv cX^{r\nu}(X^{rR} - 1) \pmod{p}.$$

The monic unit polynomial h_α has degree-preserving reduction and is not divisible by X , proving the first assertion. If the prime appears in both shapes, use $\gcd(X^a - 1, X^b - 1) = X^{\gcd(a,b)} - 1$, valid in every characteristic. $\square$

18.5 The same-shape Bezout bound

Proposition 18.8 (Same primitive shape). *Suppose distinct normalized packages have the same primitive shape $\{A, B\}$, with coprime scales r, s , and share a noncyclotomic irreducible factor of degree d . Then*

$$d \leq 2rs. \quad (18.9)$$

Proof. After replacing the common root by its inverse if necessary, put $x = \alpha^r$ and choose $y = \alpha^s$ or α^{-s} so that both x, y are roots of $Q_{A,B}$. According as the two selected orientations agree or disagree,

$$x^s = y^r \quad \text{or} \quad x^s y^r = 1.$$

Put $N = A + B$ and

$$u = \frac{A}{N}x^B, \quad 1 - u = \frac{B}{N}x^{-A}, \quad v = \frac{A}{N}y^B, \quad 1 - v = \frac{B}{N}y^{-A}.$$

Because $\gcd(A, B) = 1$, the map $x \mapsto u$ is injective on the roots under consideration: u determines both x^B and x^{-A} , and an integral Bezout combination recovers x .

In the same-orientation case the point (u, v) lies on

$$u^s = c_0 v^r, \quad (1 - u)^s = c_1 (1 - v)^r;$$

in the opposite-orientation case it lies on

$$u^s v^r = c_0, \quad (1 - u)^s (1 - v)^r = c_1,$$

with nonzero rational constants. In either case these are two divisors of bidegree (s, r) on $\mathbb{P}^1 \times \mathbb{P}^1$, whose intersection number is $2rs$.

They have no common curve component. Indeed, if one of u, v were constant on a component, either binomial equation would make the other constant as well, so the component would be a point. On the normalization of a genuine curve component, both u, v are therefore nonconstant, and the two equations give two linear relations among

$$\Delta_0 = \operatorname{div}(u), \quad \Delta_1 = \operatorname{div}(1 - u), \quad E_0 = \operatorname{div}(v), \quad E_1 = \operatorname{div}(1 - v).$$

Lemma 17.2 forces v to be one of the six transforms of u . In the basis (Δ_0, Δ_1) , their divisor pairs are

v	E_0	E_1
u	$(1, 0)$	$(0, 1)$
$1 - u$	$(0, 1)$	$(1, 0)$
u^{-1}	$(-1, 0)$	$(-1, 1)$
$(1 - u)^{-1}$	$(0, -1)$	$(1, -1)$
$u/(u - 1)$	$(1, -1)$	$(0, -1)$
$(u - 1)/u$	$(-1, 1)$	$(-1, 0)$

The same-orientation relations are $s\Delta_i = rE_i$; the table permits only $v = u$ and $r = s$, which would give the same package. The opposite-orientation relations are $s\Delta_i + rE_i = 0$, and no row satisfies both. Thus the intersection is zero-dimensional.

The d conjugates of α give d distinct intersection points. Indeed equality of the corresponding (u, v) recovers both α^r and $\alpha^{\pm s}$, and $\gcd(r, s) = 1$ recovers α . Bezout's theorem now gives (18.9). $\square$

18.6 The exact enumeration

The proof pipeline deliberately avoids constructing any large polynomial. Corollary 18.5 first enumerates all primitive shapes $A < B$, $N = A + B \leq D_1$, satisfying

$$\operatorname{rad}(A)\operatorname{rad}(B)\operatorname{rad}(N) \leq 4\,398\,937.$$

This enumeration is exhaustive without scanning every A . Order the three radicals as $\rho_1 \leq \rho_2 \leq \rho_3$. Then

$$\rho_1 \leq \lfloor 4\,398\,937^{1/3} \rfloor = 163, \quad \rho_2 \leq \lfloor 4\,398\,937^{1/2} \rfloor = 2097.$$

The program generates every integer at most D_1 with radical at most these two bounds, assigns the two smallest radical roles in all three possible ways, and reconstructs the remaining member from $A + B = N$. A canonical role ordering removes duplicates. A separate brute-force test checks this logic on small boxes.

For every coprime scale pair $1 \leq r \leq s \leq 139$, the pair program then applies, in order:

- (1) the support compatibility of Corollary 11.5 outside $\{2, 3\}$, allowing a missing prime only when it divides the opposite scale;
- (2) the two support caps (18.6), the range $D_0 \leq D \leq D_1$, and disjointness of the two scaled additive triples;
- (3) the exact integer bounds $d > D/140$, $d^3 < 64D^2$, Theorem 14.2, and the proper-factor inequalities

$$sk \leq N - 4, \quad rk \leq M - 4.$$

Here global isolation of an irreducible primitive base makes each selected primitive factor proper; Theorem 10.1 then rules out a degree-one complementary factor.

- (4) the role-binomial bounds of Lemma 18.7;
- (5) the full-radical congruences

$$sk \equiv 0, -2 \pmod{p} \quad (p \mid ABN), \quad rk \equiv 0, -2 \pmod{p} \quad (p \mid CEM),$$

combined by an exact Chinese-remainder interval test.

No irreducibility oracle, floating-point cutoff, finite-field factorization, or modular gcd is used in this main pipeline. The deterministic output is:

radical-admissible primitive shapes	237 418
scaled records	2 450 291
coprime scale pairs	5 952
shape pairs after range, support cap, and disjointness	67 730 966
after the degree interval	2 124 295
after role-binomial bounds	139 640
after full-radical CRT feasibility	761
same-shape rows among these	761
rows surviving Proposition 18.8	0.

Theorem 18.9 (Computer-assisted elimination of the unit range). *There is no normalized counterexample with*

$$6\,816\,242 \leq \max\{m, n, p, q\} \leq 175\,394\,637.$$

Consequently every normalized counterexample satisfies

$$\boxed{\max\{m, n, p, q\} \leq 6\,816\,241.} \tag{18.10}$$

Proof. Every counterexample in the displayed range satisfies all of the mathematical filters just listed, so the exhaustive exact program places it among the 761 final rows. In every row the two primitive shapes agree. Proposition 18.8 gives

$$d \leq 2rs \leq 2 \cdot 139 \cdot 138 = 38\,364,$$

whereas Proposition 18.2 gives $d \geq 48\,688$. Thus all rows are impossible. Combining this with Theorem 17.13 proves (18.10). $\square$

Remark 18.10 (Certificate boundary). The theorem is computer-assisted in the ordinary source-verification sense. The bundle contains the complete shape list, the 761 pre-final rows, an independent Python verifier for every recorded row, deterministic checksums, and a brute-force small-box regression test for the shape enumerator. The coverage of the large pair scan is encoded transparently in the short C++ loop rather than in a proof-assistant certificate. A future Lean-oriented version should replace that last trust boundary by interval or branch coverage certificates; no large polynomial arithmetic would be required.

18.7 Certified constants and reproduction commands

The main command is

```
make upper THREADS=25
make verify
make test
```

The proof pipeline requires a C++20 compiler with OpenMP and Python 3; its Python verification uses only the standard library. The exact rational certificate `code/certify_constants.py` prints

```
plastic_power_coefficients=62608681,109870576,82938844
theta70_rational_lower=220094051941/625
twice_D_max=350789274
uniform_degree_bound=d>D/140
scale_max=139
same_shape_max=38364
unit_degree_integer_min=48688
support_endpoint_upper_floor=4398936
support_product_cap=4398937
```

The two analytic transition points in Theorem 17.13 and Corollary 17.14 are independently checked by the supplied Sage and Arb script `code/certify_thresholds.sage`. All comparisons in both scripts are interval or exact rational comparisons, not binary floating-point guesses.

18.8 Exact factorization through total sixty

Exact factorization over $\mathbb{Q}$ was rerun with SymPy 1.14 for every unordered unequal pair with total $a + b \leq 60$. There are

$$\sum_{N=3}^{60} \left\lfloor \frac{N-1}{2} \right\rfloor = 870$$

such pairs. Every $Q_{a,b}$ was irreducible, and no normalized reciprocal factor orbit occurred in two different packages. The calculation took exact quotients and exact rational factorization; no numerical root matching was used.

For every tested pair, its primitive reduction also has total at most 60 and was found irreducible. Theorem 14.9 therefore upgrades all 870 finite factorizations to global certificates: every one of these packages is coprime to every distinct package in the entire infinite family, including packages of arbitrarily large total.

The complete script is included below.

```
from math import gcd
import sympy as sp
from sympy import Poly, QQ

x = sp.symbols("x")
MAXN = 60
seen, reducible, collisions = {}, [], []
count = globally_certified = 0

def qpoly(a, b):
    g = gcd(a, b)
    num = Poly(a*x**(a+b) - (a+b)*x**a + b, x, domain=QQ)
    den = Poly((x*g - 1)**2, x, domain=QQ)
    return num.exquo(den)

def reciprocal_orbit_key(f):
    p = Poly(f, x, domain=QQ).monic()
    rev = Poly.from_list(list(reversed(p.all_coeffs()))),
    gens=x, domain=QQ).monic()
    return min(tuple(p.all_coeffs()), tuple(rev.all_coeffs()))

for N in range(3, MAXN + 1):
    for a in range(1, (N - 1)//2 + 1):
        b = N - a
        count += 1
        q = qpoly(a, b)
        _, factors = sp.factor_list(q.as_expr(), x, domain=QQ)
        is_irreducible = (len(factors) == 1 and factors[0][1] == 1
```

```

        and Poly(factors[0][0], x).degree() == q.degree())
if not is_irreducible:
    reducible.append((a, b))
g = gcd(a, b)
base = qpoly(a//g, b//g)
_, base_factors = sp.factor_list(base.as_expr(), x, domain=QQ)
base_irreducible = (len(base_factors) == 1
                    and base_factors[0][1] == 1
                    and Poly(base_factors[0][0], x).degree() == base.degree())
if base_irreducible:
    globally_certified += 1
for f, exponent in factors:
    key = reciprocal_orbit_key(f)
    if key in seen and seen[key] != (a, b):
        collisions.append((seen[key], (a, b)))
    else:
        seen[key] = (a, b)

print(count, reducible, collisions, globally_certified)
# Output: 870 [] [] 870

```

19 Lower-range extraction degrees and endpoint states

The full conjecture is not yet proved. The audited upper computation gives the exact reduction

$$D := \max\{m, n, p, q\} \leq 6\,816\,241.$$

The obstruction to simply extending that computation is that the common minimal polynomial need not be monic with unit constant term. Full Kummer degree and the one-package unit support cap may therefore fail at endpoint primes. This section replaces full Kummer degree by exact extraction degrees and gives a finite state model for every remaining branch.

Write the two normalized packages as

$$\{ra, rb\}, \quad \{sc, se\}, \quad \gcd(a, b) = \gcd(c, e) = 1, \quad \gcd(r, s) = 1,$$

and let α be a common root of selected orientations. Put

$$E = \mathbb{Q}(\alpha), \quad K = \mathbb{Q}(\alpha^r), \quad L = \mathbb{Q}(\alpha^s),$$

$$d = [E : \mathbb{Q}], \quad e_1 = [K : \mathbb{Q}], \quad e_2 = [L : \mathbb{Q}], \quad t = [E : K], \quad u = [E : L].$$

19.1 Degree bookkeeping without full Kummer degree

Lemma 19.1 (Extraction-degree identities). *One has*

$$1 \leq t \leq r, \quad 1 \leq u \leq s, \quad d = te_1 = ue_2, \quad d \leq e_1e_2. \quad (19.1)$$

Moreover $E = KL$.

Proof. The element α satisfies $X^r - \alpha^r$ over K and $X^s - \alpha^s$ over L , proving the first two inequalities and the tower identities. Since $\gcd(r, s) = 1$, an integral Bézout combination of r, s recovers α from α^r, α^s , so $E = KL$. The compositum inequality $[KL : \mathbb{Q}] \leq [K : \mathbb{Q}][L : \mathbb{Q}]$ gives the last assertion. $\square$

The last inequality is intentionally weaker than a linear-disjointness identity. No such identity is assumed in the lower-range state model.

19.2 A reduced power-correspondence bound

Proposition 19.2 (General reduced correspondence). *Let selected primitive orientations be $Q_{a,b}(x^r)$ and $Q_{c,e}(x^s)$, and suppose they share a noncyclotomic irreducible factor of degree d . Put*

$$g_0 = \gcd(rb, se), \quad B_0 = \frac{rb}{g_0}, \quad E_0 = \frac{se}{g_0},$$

$$g_1 = \gcd(ra, sc), \quad A_0 = \frac{ra}{g_1}, \quad C_0 = \frac{sc}{g_1}.$$

Unless the normalized packages are identical,

$$d \leq E_0 A_0 + B_0 C_0 = \frac{rs(ae + bc)}{\gcd(rb, se) \gcd(ra, sc)}. \quad (19.2)$$

Proof. Set $x = \alpha^r$, $y = \alpha^s$, $N = a + b$, $M = c + e$, and

$$U_0 = \frac{a}{N} x^b, \quad 1 - U_0 = \frac{b}{N} x^{-a},$$

$$V_0 = \frac{c}{M} y^e, \quad 1 - V_0 = \frac{e}{M} y^{-c}.$$

The relation $x^s = y^r$ gives

$$U_0^{E_0} = \lambda_0 V_0^{B_0}, \quad (1 - U_0)^{C_0} = \lambda_1 (1 - V_0)^{A_0}, \quad (19.3)$$

with nonzero rational constants. Their closures in $\mathbb{P}^1 \times \mathbb{P}^1$ have bidegrees (E_0, B_0) and (C_0, A_0) , so their intersection number is $E_0 A_0 + B_0 C_0$.

They have no common curve component unless the packages agree. On the normalization of a hypothetical component, the two equations give two linear relations among the four divisors of $U_0, 1 - U_0, V_0, 1 - V_0$. Lemma 17.2 makes V_0 one of the six Möbius transforms of U_0 . In the divisor table from the proof of Proposition 18.8, positivity in

$$E_0 \operatorname{div}(U_0) = B_0 \operatorname{div}(V_0), \quad C_0 \operatorname{div}(1 - U_0) = A_0 \operatorname{div}(1 - V_0)$$

rules out five rows. The remaining row is $V_0 = U_0$, with $E_0 = B_0$ and $C_0 = A_0$. Thus $se = rb$ and $sc = ra$; primitivity gives $(c, e) = (a, b)$ and $r = s$.

The map $x \mapsto U_0$ is injective on the relevant roots, because U_0 determines both x^b and x^{-a} , and $\gcd(a, b) = 1$. Similarly V_0 determines y . The d conjugates of α therefore give d distinct intersection points: equality of the two coordinates recovers α^r, α^s , and then α . Bézout proves (19.2). $\square$

For equal primitive shapes, $g_0 = b$ and $g_1 = a$, so (19.2) recovers $d \leq 2rs$.

19.3 Scaled coefficient arithmetic

Let

$$h_\alpha(X) = UX^d + \cdots + V \in \mathbb{Z}[X], \quad U > 0,$$

be the primitive minimal polynomial of α , and let f_β, f_γ be the primitive minimal polynomials of $\beta = \alpha^r, \gamma = \alpha^s$, with leading and constant coefficients $(u_1, v_1), (u_2, v_2)$.

Lemma 19.3 (Norm transfer). *With the notation above,*

$$u_1^t = U^r, \quad |v_1|^t = |V|^r, \quad u_2^u = U^s, \quad |v_2|^u = |V|^s. \quad (19.4)$$

In particular, the prime support of each pair (u_i, v_i) equals the corresponding part of the prime support of UV .

Proof. The norm identity

$$N_{E/\mathbb{Q}}(\alpha^r) = N_{K/\mathbb{Q}}(\beta)^t$$

reads $(V/U)^r = (v_1/u_1)^t$, up to sign. Both fractions are in lowest terms, so numerator and denominator valuations agree prime by prime. The second package is identical. $\square$

Proposition 19.4 (Scaled endpoint divisibilities and slopes). *For the first selected orientation,*

$$|V|^{ra} \mid b^d, \quad U^{rb} \mid a^d; \quad (19.5)$$

for the second,

$$|V|^{sc} \mid e^d, \quad U^{se} \mid c^d. \quad (19.6)$$

If $p \mid U$, then

$$\frac{v_p(a)}{rb} = \frac{v_p(c)}{se}; \quad (19.7)$$

and if $q \mid V$, then

$$\frac{v_q(b)}{ra} = \frac{v_q(e)}{sc}. \quad (19.8)$$

Proof. Apply Proposition 17.20 to f_β and raise its endpoint divisibilities to the power t . Lemma 19.3 and $te_1 = d$ give (19.5). The second package gives (19.6). For (19.7), choose a p -adic embedding at which α is a negative-valuation root. The endpoint Newton polygons give respectively $-v_p(a)/(rb)$ and $-v_p(c)/(se)$. The positive-valuation argument gives (19.8). $\square$

Proposition 19.5 (Two nonunit coefficients lie in a small box). *If $U > 1$ and $|V| > 1$, then*

$$D \leq 37\,799. \quad (19.9)$$

Proof. Choose distinct primes $p \mid U$, $q \mid V$; they are distinct because $\gcd(U, V) = 1$. Equation (19.7) implies

$$B_0 \leq v_p(a), \quad E_0 \leq v_p(c),$$

and (19.8) gives

$$A_0 \leq v_q(b), \quad C_0 \leq v_q(e).$$

Proposition 19.2 therefore yields

$$d \leq v_p(c)v_q(b) + v_p(a)v_q(e) \leq 2\lfloor \log_2 D \rfloor \lfloor \log_3 D \rfloor.$$

Together with $d > D/140$, this is impossible for $D \geq 37\,800$. The exact integer transition is certified by `code/lower/certify_lower_constants.py`. $\square$

Corollary 19.6 (Coefficient trichotomy above 50 000). *If $50\,000 \leq D \leq 6\,816\,241$, then*

$$M(h_\alpha) < 13.$$

Consequently either $(U, |V|) = (1, 1)$, or exactly one of $U, |V|$ is a nonunit integer in $\{2, 3, \dots, 12\}$.

Proof. The height–degree estimate used in Section 18 gives

$$\log M(h_\alpha) < \frac{8 \log(2D)}{D^{1/3}}.$$

The right side is decreasing for $D \geq 50\,000$. The exact rational interval certificate accompanying this revision proves that its value at 50 000 is below $\log 13$. The leading and constant coefficients of a primitive polynomial are at most its Mahler measure. Proposition 19.5 excludes two nonunit coefficients in this range. $\square$

19.4 Deficient scale primes

Proposition 19.7 (Localization of a Kummer defect). *Let $p^a \parallel r$. If $v_p(t) < a$, then either $\beta \in K^p$, or $p = 2$, $a \geq 2$, and $\beta \in -4K^4$.*

The following alternatives exhaust the resulting local branch.

(a) *If p is good for the primitive shape, then*

$$e_1 \leq p - 2. \tag{19.10}$$

(b) *A total-role prime is excluded by Corollary 15.5.*

(c) *An endpoint-role prime is excluded when f_β is a unit polynomial. In the remaining nonunit endpoint packet, writing $p^\kappa \parallel a$ when the leading coefficient is nonunit, one has*

$$e_1 \leq b, \quad pb \mid e_1 \kappa. \tag{19.11}$$

The reciprocal statement holds when the constant coefficient is nonunit. Moreover $p \in \{2, 3, 5, 7\}$.

In particular every unit Kummer-defect counterexample satisfies

$$D < 140 \cdot 139 \cdot 137 = 2\,666\,020. \tag{19.12}$$

Proof. If $\beta \notin K^p$ and the exceptional fourth-power condition does not occur, Capelli’s criterion makes $X^{p^a} - \beta$ irreducible. The field $K(\alpha^{r/p^a})$ is then a subfield of E of degree p^a , so $p^a \mid t$, a contradiction.

The good-prime alternative is Theorem 14.5; the total-role exclusion is Corollary 15.5. At an endpoint, Theorem 15.6 excludes a unit factor and gives (19.11) for the nonunit packet, including the exceptional fourth-power branch at $p = 2$. Its necessary condition $\kappa \geq p$, together with $p^\kappa \leq D \leq 6\,816\,241$, leaves $p = 2, 3, 5, 7$.

For a unit factor every deficient prime is therefore good and satisfies $e_1 \leq p - 2$. Since $d = te_1 \leq re_1$, $r \leq 139$, and $p \leq 139$, the inequality $d > D/140$ gives (19.12). $\square$

19.5 A finite certificate object

Definition 19.8 (Lower endpoint state). A one-package state is a tuple

$$(r, t, e_1, d, U, V; a, b, \varepsilon),$$

where ε records the selected orientation. A two-package state consists of two such tuples with a common (d, U, V) . Put

$$N = a + b, \quad M = c + e, \quad D = \max(rN, sM).$$

It is *admissible* if all of the following hold:

- (i) $1 \leq t \leq r \leq 139$, $d = te_1$, $2 \leq e_1 \leq N - 4$, and analogously on the second side;
- (ii) $140d > D$, $d^3 < 64D^2$, and $d \leq \min(N, M) - 2$;
- (iii) for every coefficient prime ℓ ,
$$t \mid rv_\ell(U), \quad t \mid rv_\ell(V),$$
and analogously for (s, u, e_2) ;
- (iv) the scaled endpoint divisibilities (19.5)–(19.6), the slope equalities (19.7)–(19.8), and the correspondence bound (19.2);
- (v) every prime in the primitive additive triple outside $\text{Supp}(UV)$ satisfies the factor-degree alternatives of Theorem 15.1, Proposition 15.10, and Corollary 15.11;
- (vi) every deficient scale prime satisfies one of the three alternatives in Proposition 19.7;
- (vii) the scaled additive triples are disjoint, the scales are coprime, and the two packages are distinct.

Theorem 19.9 (Finite lower-state envelope). *Every normalized counterexample with $D \leq 6\,816\,241$ determines an admissible lower endpoint state. The set of such states is finite and effectively enumerable. Consequently, an exhaustive elimination of the admissible states proves the conjecture.*

Proof. Lemmas 19.1 and 19.3, Propositions 19.2, 19.4, and 19.7, and the exact local factor signatures of Section 15 give the listed necessary conditions. All integer entries are explicitly bounded by $D \leq 6\,816\,241$, $r, s \leq 139$, the degree bounds, and the endpoint divisibilities. Thus the state set is finite and can be enumerated with integer arithmetic. The last sentence is the contrapositive of the first. $\square$

19.6 Computational implementation and present assurance boundary

The accompanying bundle separates four layers:

- (1) the independently audited upper-range program and deterministic data;
- (2) exact lower constants, a machine-readable endpoint-state schema, and a reference checker which recomputes all global arithmetic constraints;
- (3) a literal small-domain endpoint-state enumerator, used for coverage regressions independently of a future optimized production generator;
- (4) terminal finite-field certificates. A certificate records a good auxiliary prime for which the two collision trinomials preserve degree and have modular gcd of degree exactly two; a separate checker recomputes that claim.

Development runs found no terminal survivor in the implemented branches and indicate that arithmetic complexity is comfortably below the stated 25-core, seven-day budget. Those observations are not promoted to a theorem in this revision. The remaining assurance obligation is to consolidate the development enumerators into one deterministic production generator, emit a branch-coverage manifest, and compare it with a structurally independent coverage implementation. The exact checklist is in `docs/lower/PROOF_BOUNDARY.md`.

Conjecture 19.10 (Residual four-index lemma). *For $|z| \neq 1$,*

$$L_m(z) = L_n(z), \quad L_p(z) = L_q(z), \quad m < n, \quad p < q,$$

implies $\{m, n\} = \{p, q\}$.

Conclusion

The audited upper computation excludes

$$6\,816\,242 \leq D \leq 175\,394\,637.$$

The present revision supplies the mathematical architecture needed below that range without assuming full Kummer degree. It proves the general reduced-correspondence bound, transfers endpoint coefficients through exact norm identities, confines the two-nonunit branch to $D \leq 37\,799$, confines every unit Kummer-defect branch to $D < 2\,666\,020$, and reduces the remaining nonunit defects to endpoint packets at $p \in \{2, 3, 5, 7\}$.

This is a finite and realistically sized computer-assisted proof program, but the package does not yet declare the conjecture proved. Its outstanding obligation is now explicit and software-theoretic: a production lower-state manifest and an independent coverage audit, followed by verification of the terminal modular certificates. Keeping that boundary visible is essential for the next independent review.

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